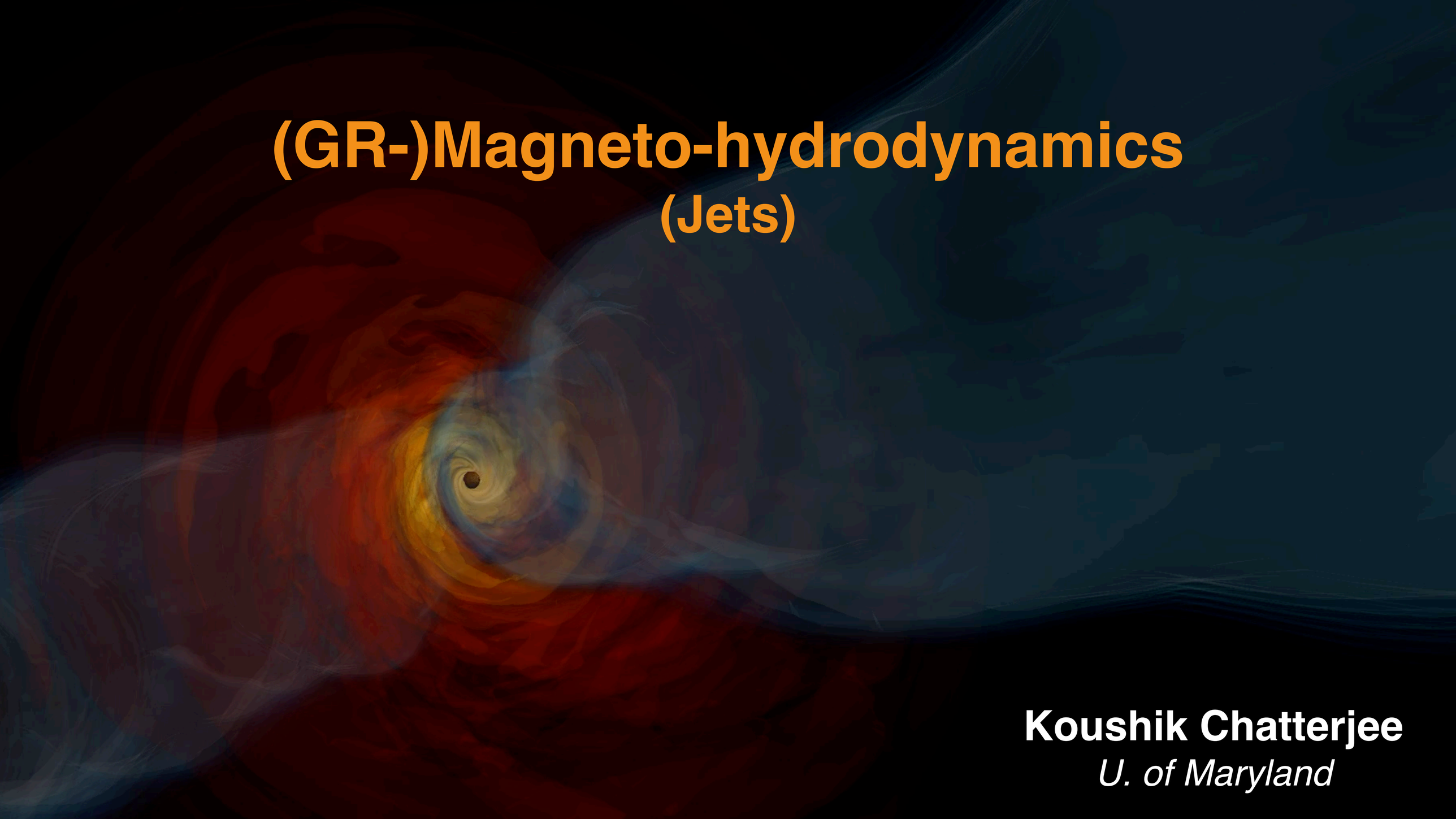
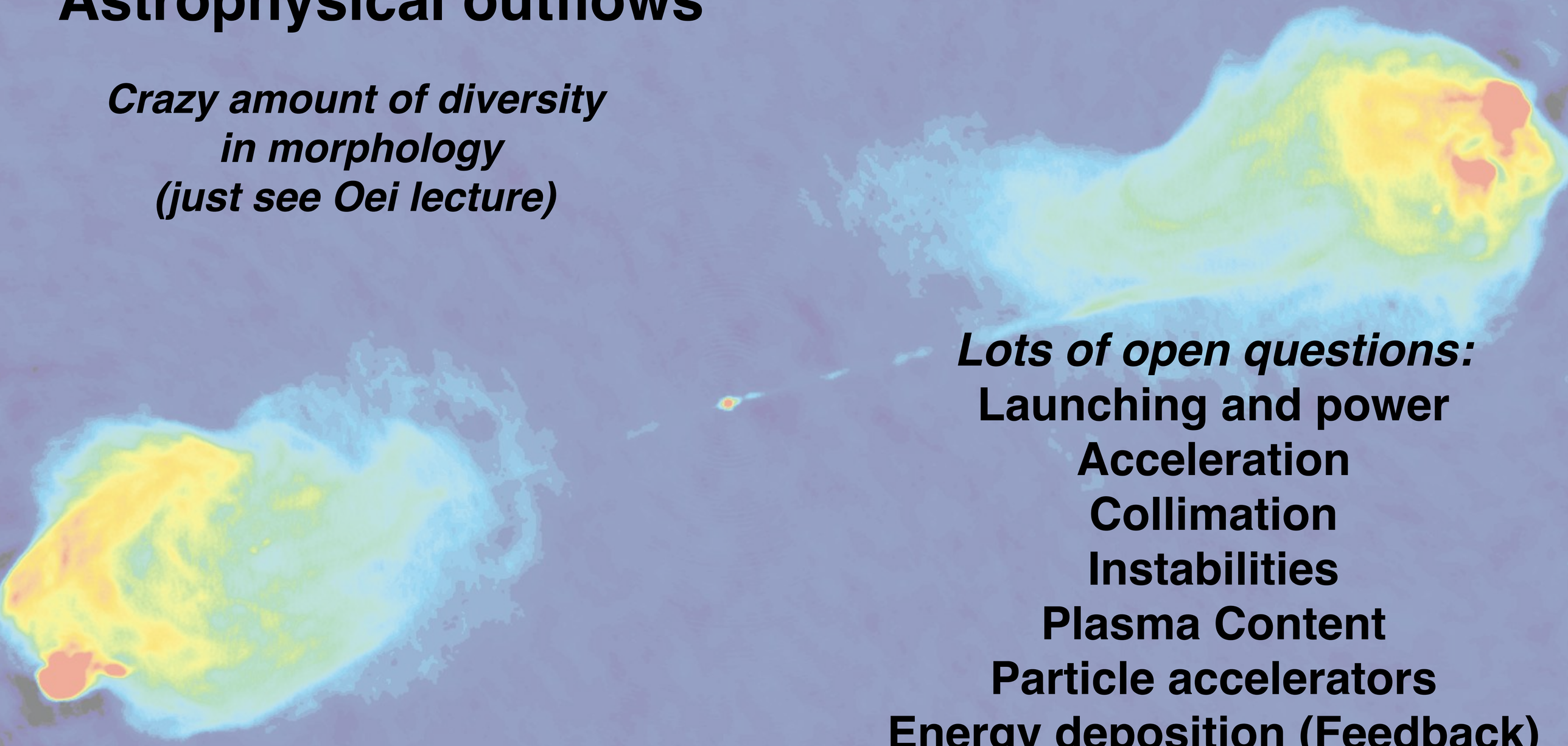


# (GR-)Magneto-hydrodynamics
## (Jets)

**Koushik Chatterjee**
*U. of Maryland*

# Astrophysical outflows

***Crazy amount of diversity in morphology (just see Oei lecture)***

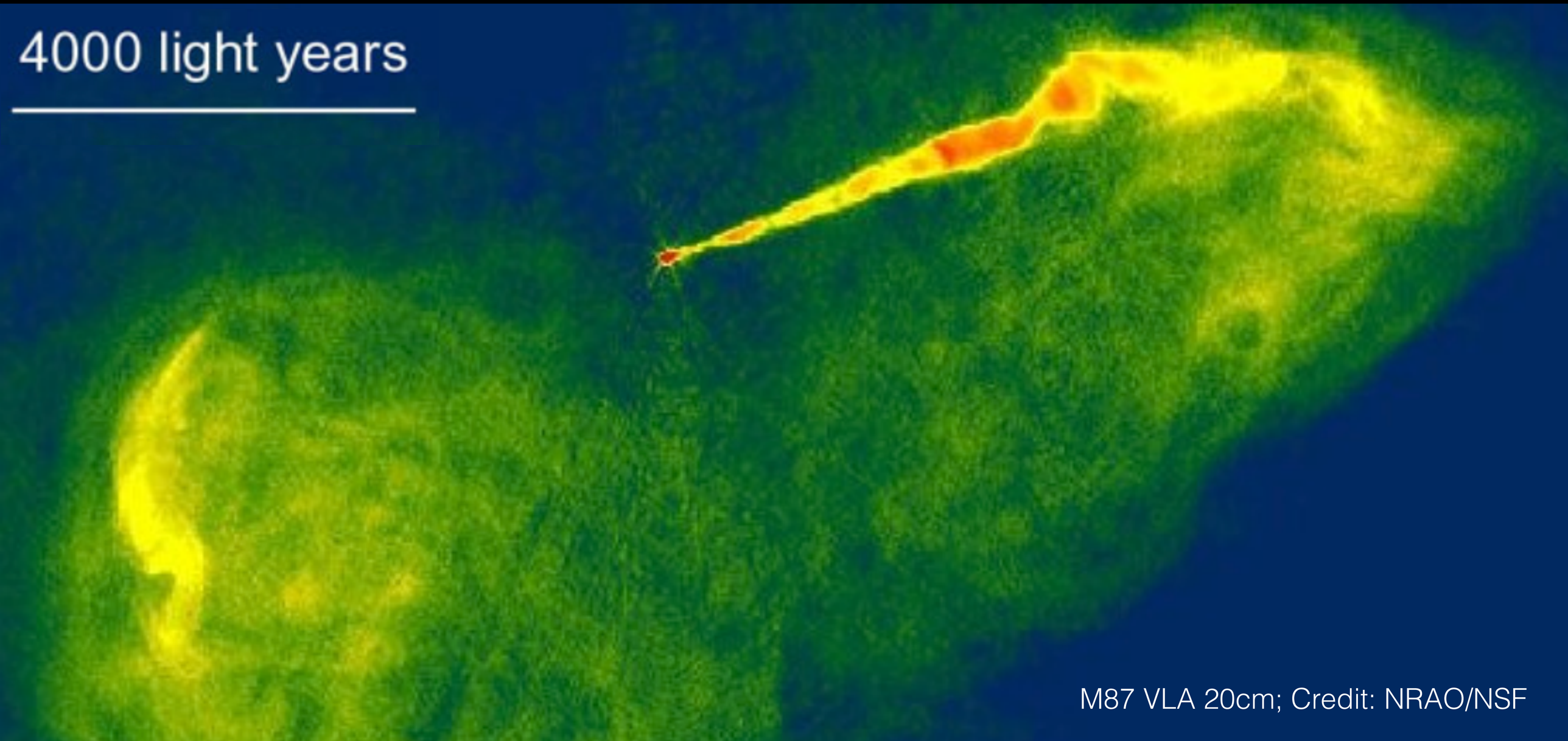


***Lots of open questions:***

**Launching and power**
**Acceleration**
**Collimation**
**Instabilities**
**Plasma Content**
**Particle accelerators**
**Energy deposition (Feedback)**

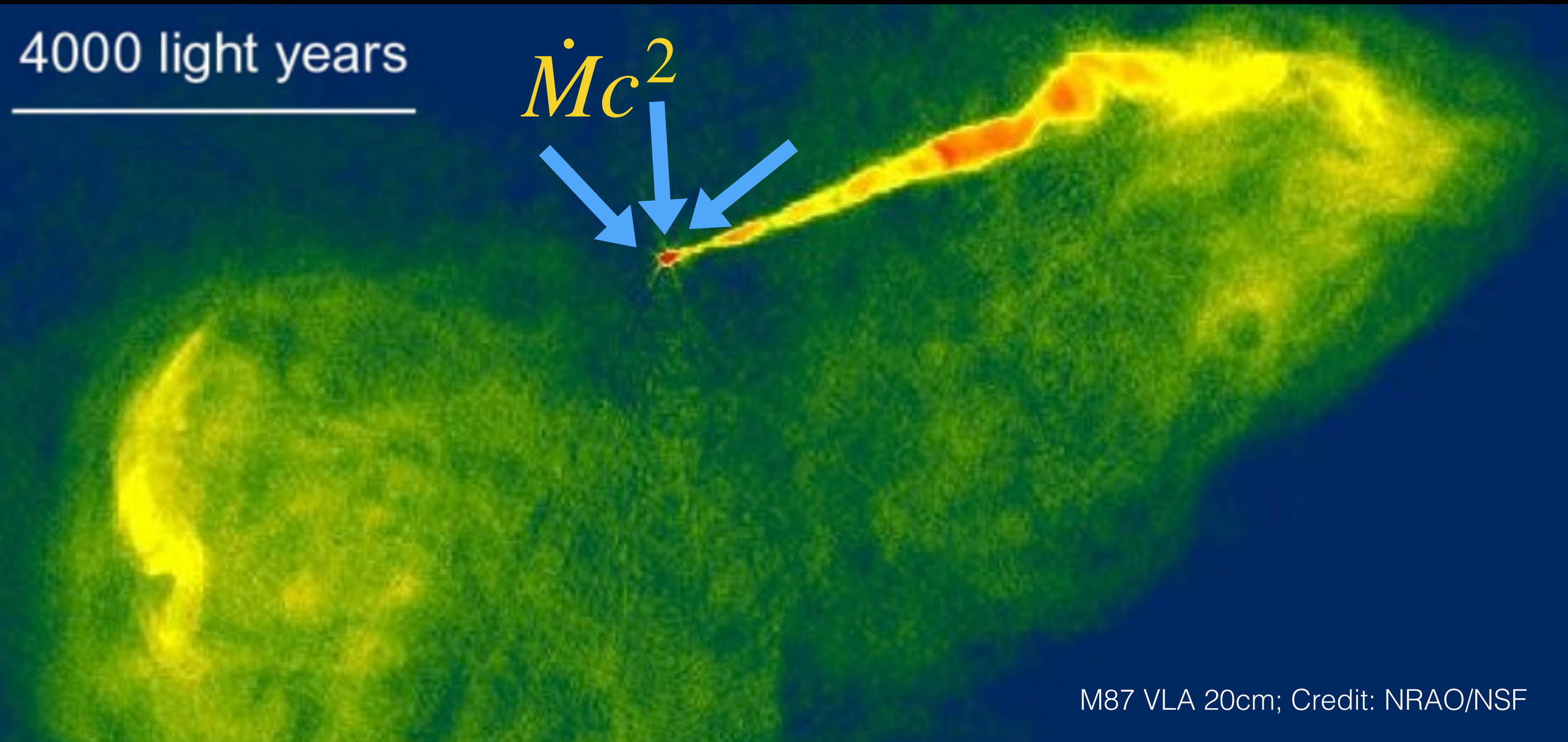
CYGNUS A: Image courtesy of NRAO/AUI; R. Perley, C. Carilli & J. Dreher

# Jets in nature: M87



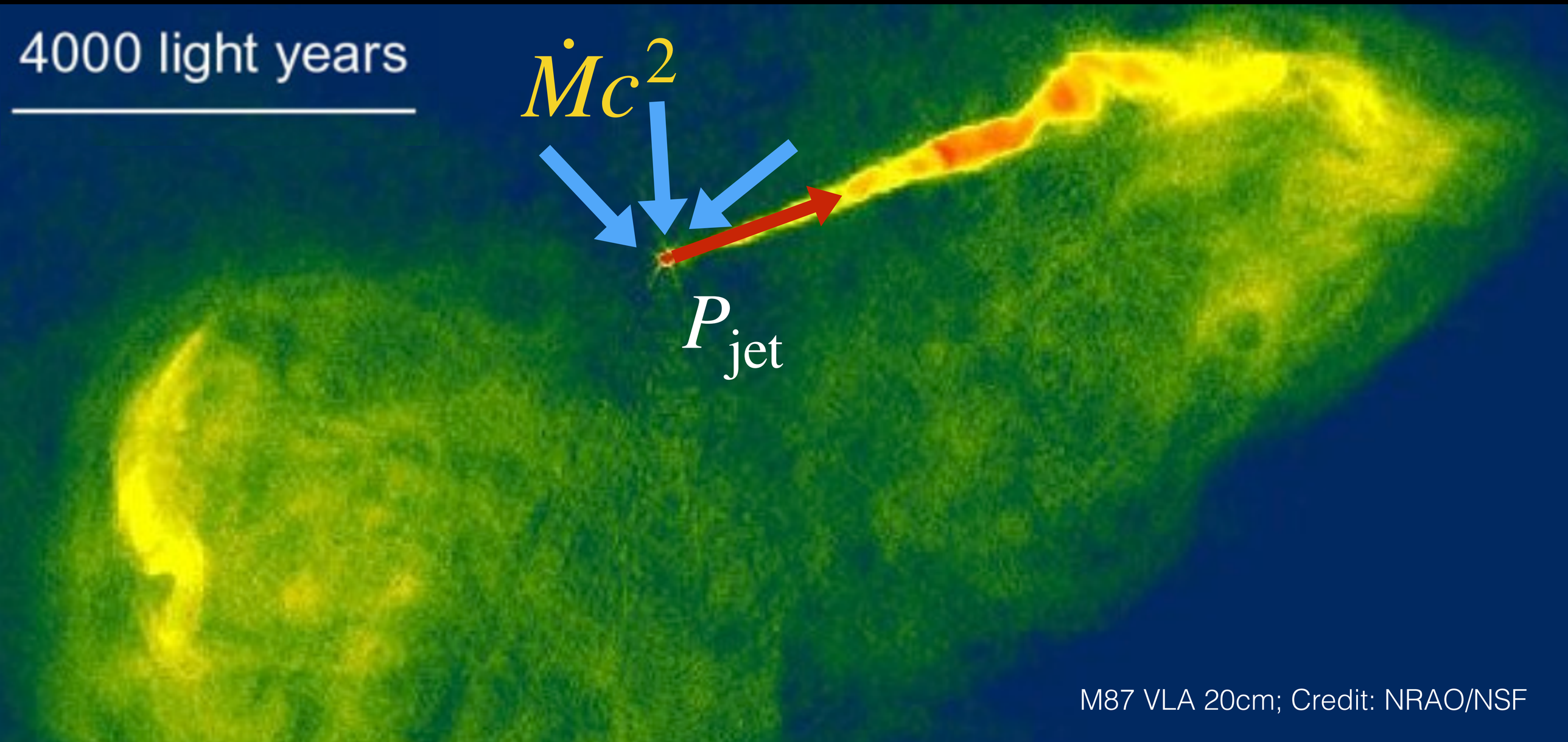

M87 VLA 20cm; Credit: NRAO/NSF

# Jets in nature: M87



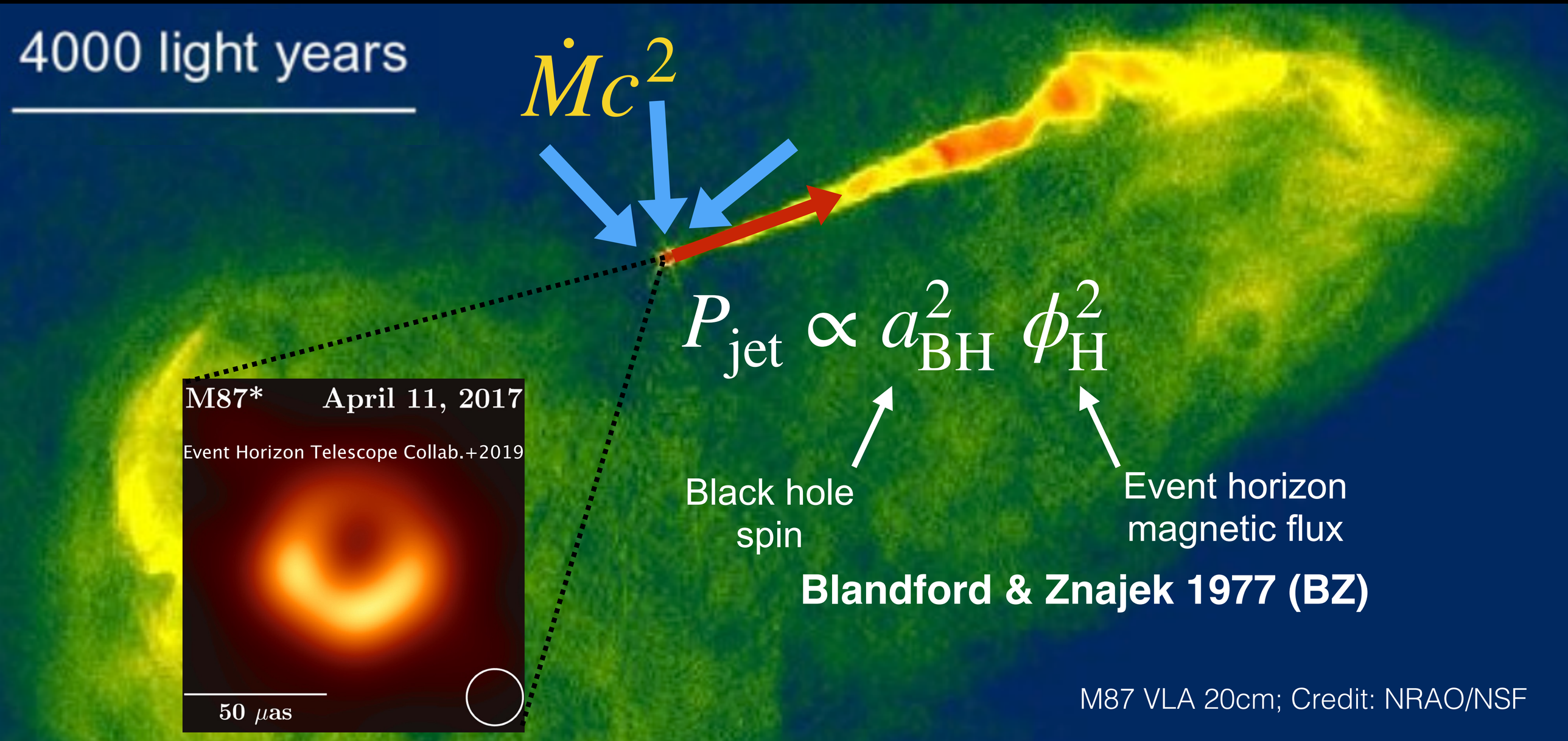

M87 VLA 20cm; Credit: NRAO/NSF

# Jets in nature: M87




M87 VLA 20cm; Credit: NRAO/NSF

# Jets: Spin Energy Extraction



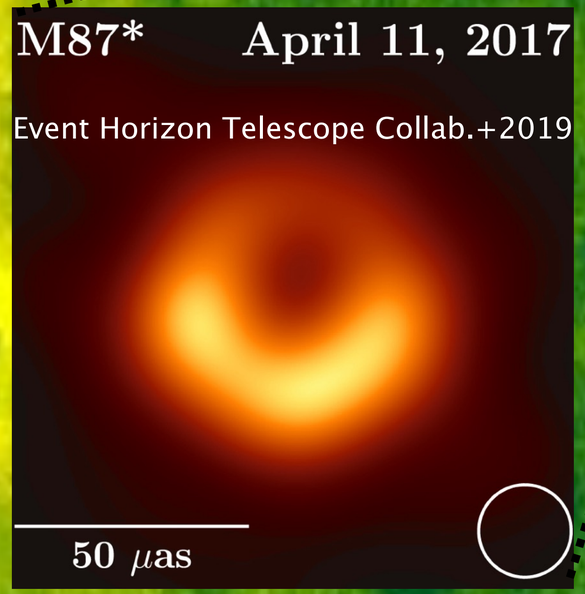
# Jets: A Multi-scale Problem

4000 light years

(~4.1x10$^6$ GM/c$^2$ or $r_g$)

$\dot{M}c^2$

$P_{\rm jet} \propto a_{\rm BH}^2 \, \phi_{\rm H}^2$



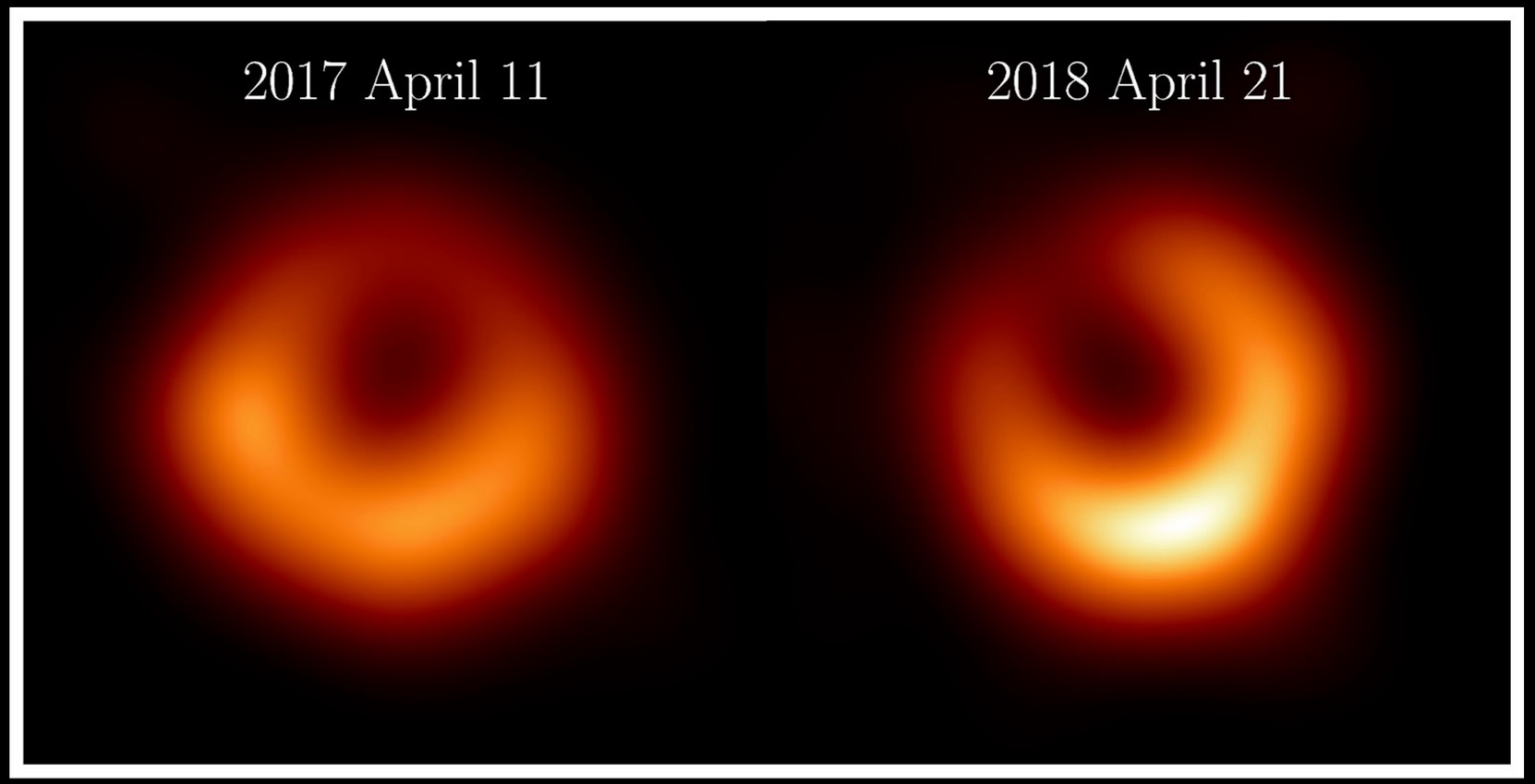

~7-8 orders of separation in length-scale!!

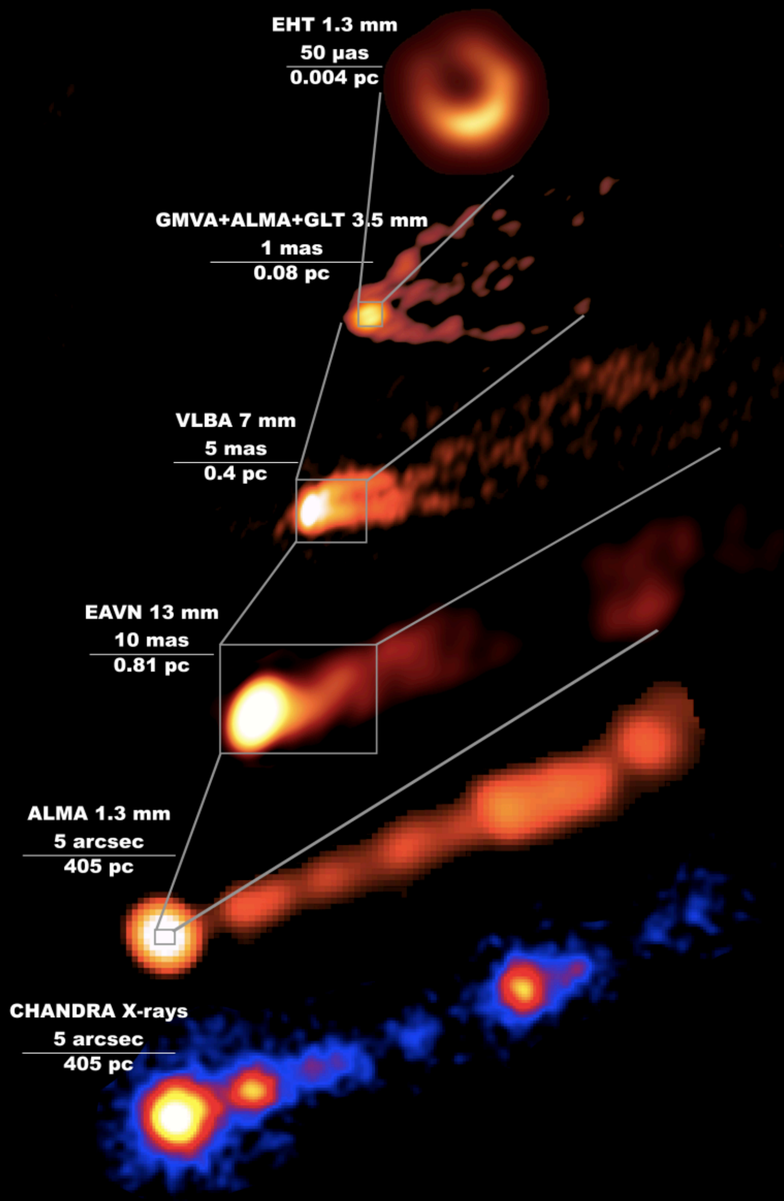
(Angles-Alcazar+, Cho+, Guo+, Hopkins+, KC+, Lalakos+, Ressler+, many)

M87 VLA 20cm; Credit: NRAO/NSF

# Multi-Year Observations of A Black Hole

EHT M87 2017, 2018 (EHTC+19,+24)




M87 2017, 2018 Multi-wavelength (MWL) campaigns: Algaba+21,+24

# Wind Solutions from (M)HD

E.g., Parker solar wind: steady, spherically symmetric outflow applied to the Sun's corona (Hydrodynamic solution)

$$\left(v^2 - c_s^2\right) \frac{dv}{dr} = v \left( \frac{2c_s^2}{r} - \frac{GM}{r^2} \right)$$

Critical points: where both numerator and denominator vanish in dv/dr=N/D

Here that happens when fluid velocity equals the local sound speed at a critical radius $r_c = GM/2c_s^2$

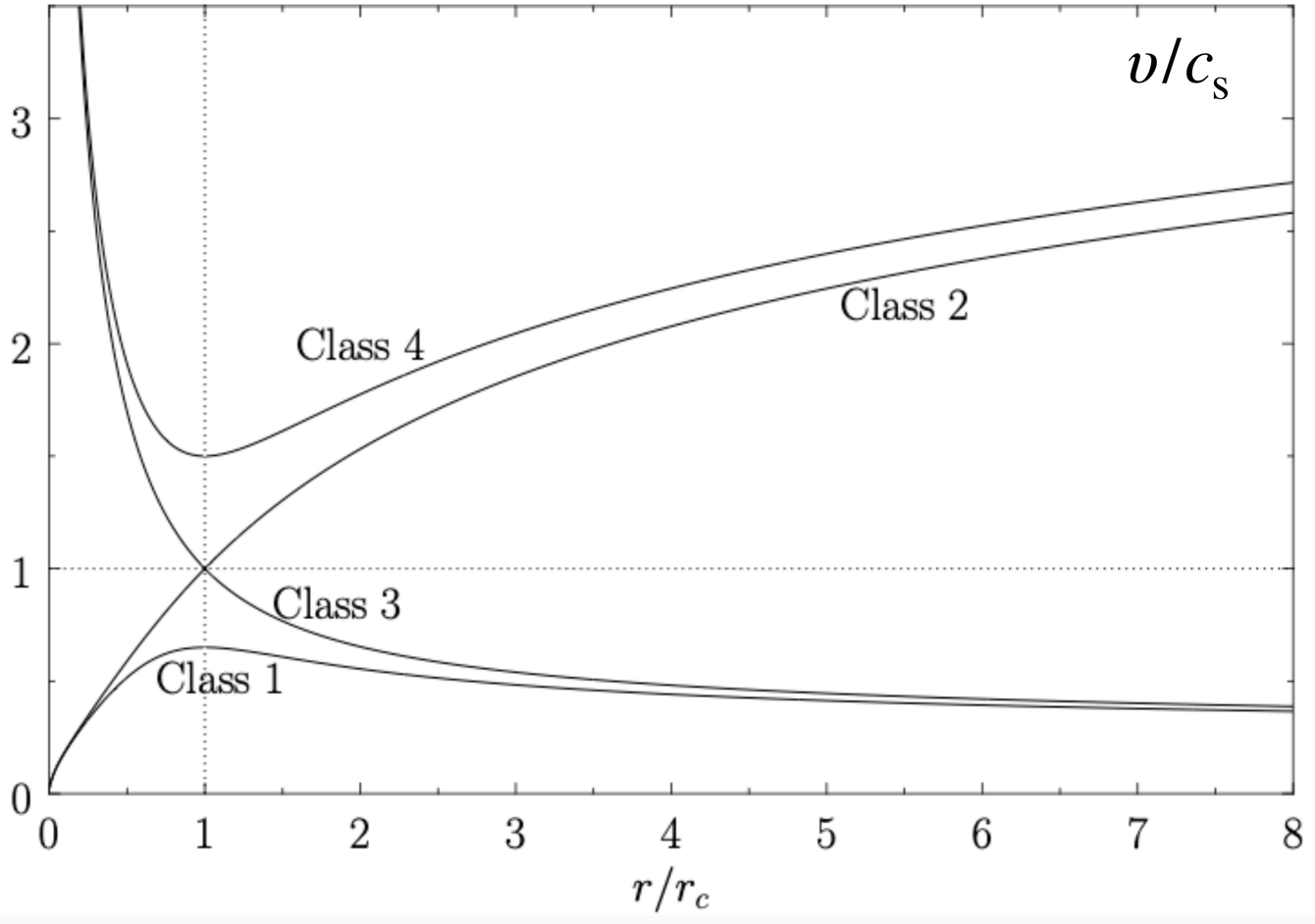
This is a transonic flow, where the fluid accelerates with radius and becomes supersonic beyond the critical radius

Magnetized winds: Weber-Davis solution shows that the critical points are related to the Alfven speed: $v_r = v_{\mathrm{A}}$

# Wind Solutions from (M)HD

E.g., Parker solar wind: steady, spherically symmetric outflow applied to the Sun's corona (Hydrodynamic solution)

Solutions depend on boundary conditions

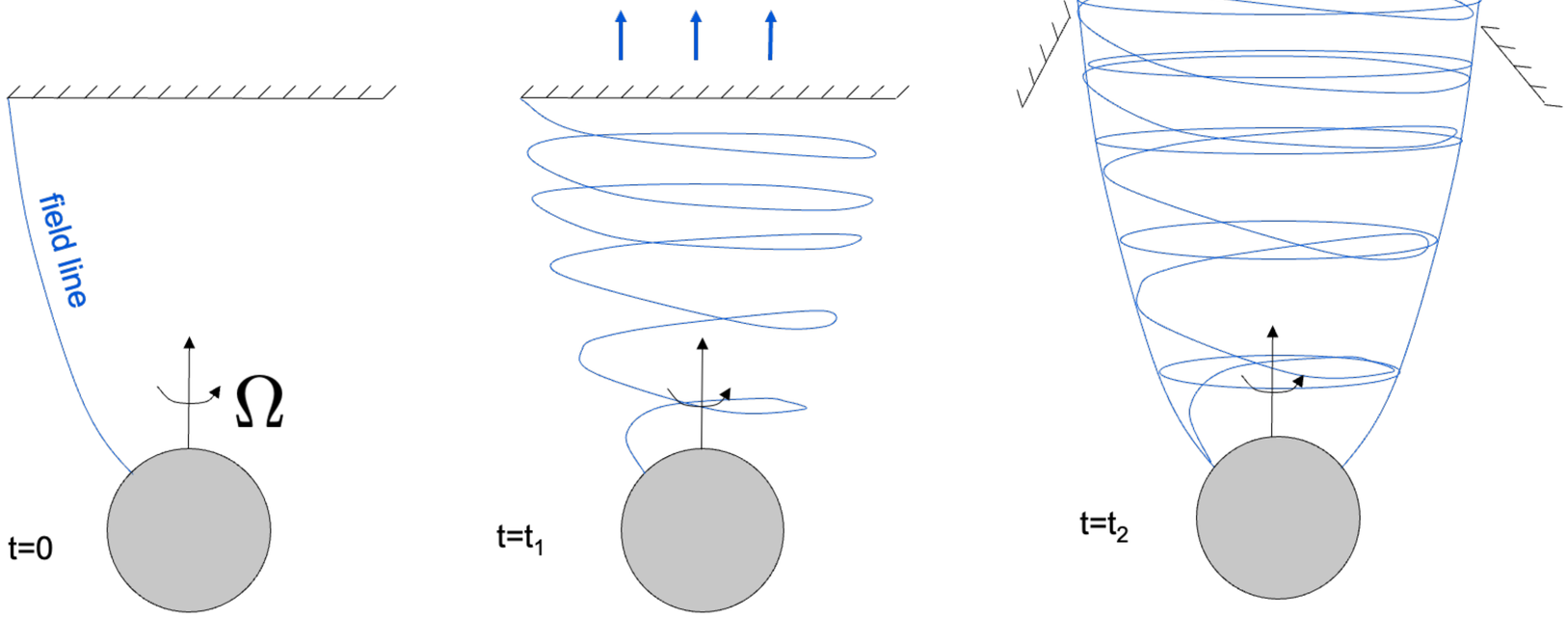


# Rotation-powered outflows: launching

$$P = \frac{B_\varphi^2}{8\pi}$$

Field toroidally-dominated

$$B_\varphi \gg B_z$$




(Beskin & Nokhrina 06, Komissarov 07-10, AT+08,09,10,12, Lyubarsky 10, AT 15)

Credit: Sasha Tchekhovskoy

# Powering outflows: Poynting flux

Poynting flux $P \sim \int_S \frac{1}{4\pi}(\vec{E} \times \vec{B}) \,.\, dA$, calculated over a surface of radius $r$

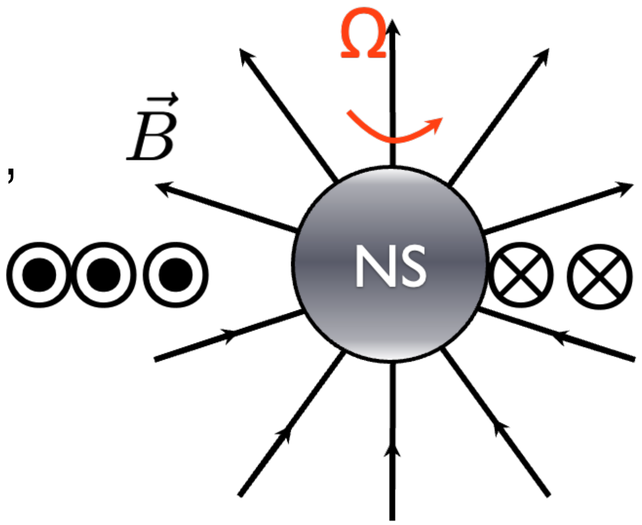
Michel 69,73

Here, field lines are twisted via rotation: $|B_\varphi| \sim \Omega R B_r \sim E_\theta \qquad (R \equiv r \sin\theta)$

$$P = \int \frac{1}{4\pi}(\Omega r \sin\theta B_r)^2 dA \sim \frac{1}{4\pi}(\Omega r B_r)^2 (2\pi r^2) \int (\sin\theta)^3 d\theta = \frac{2}{3}\Omega^2 r^4 B_r^2$$

Magnetic flux $\Phi \equiv (1/2)\int B \cdot dA = 2\pi r^2 |B_r|$,

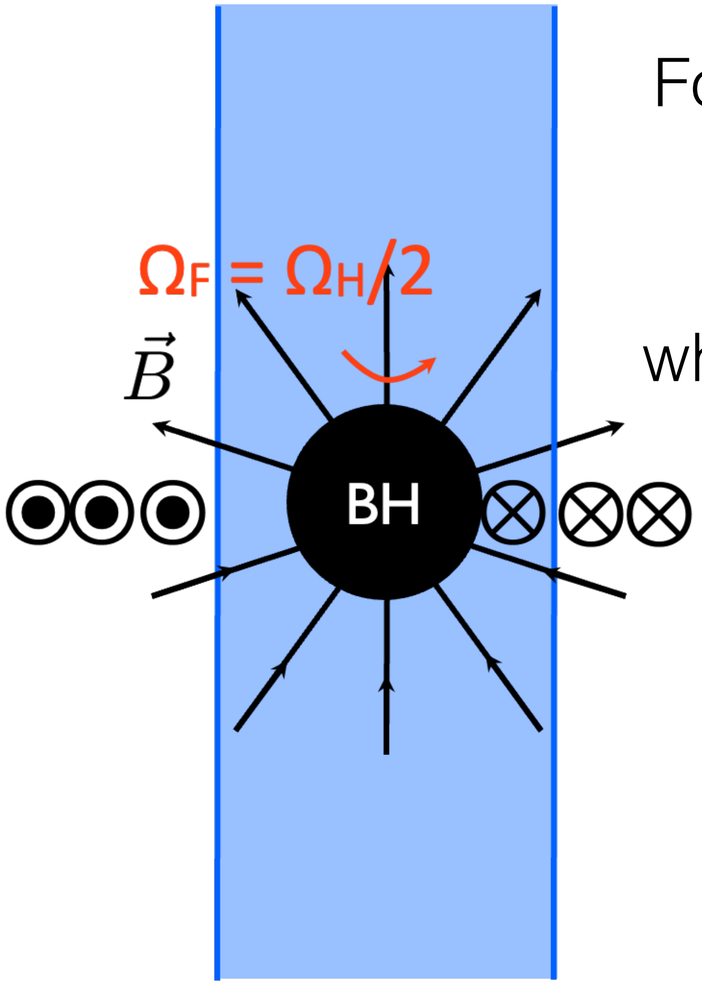
therefore, $P = \frac{1}{6\pi^2}\Phi^2\Omega^2$




$\Omega = 1/R_{\rm LC}$

Credit: Sasha Tchekhovskoy

# Powering outflows: Poynting flux




For black holes, $\Omega \to \Omega_{\rm F}$, the field line angular velocity

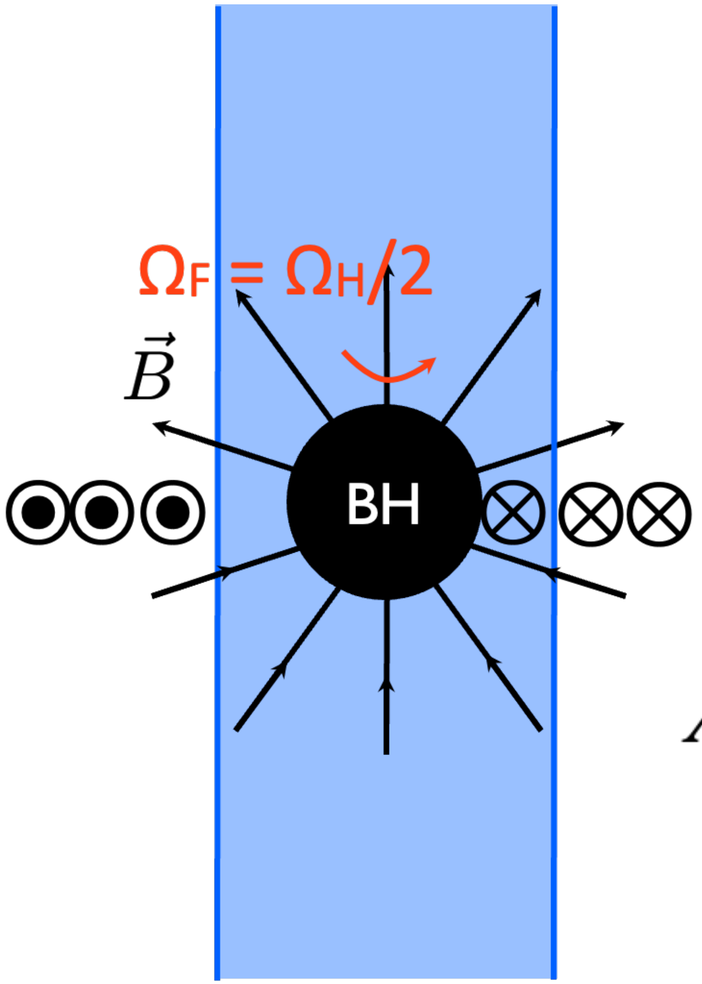
Empirically, it has been found that $\Omega_{\rm F} \approx \Omega_{\rm H}/2$,
where $\Omega_{\rm H} = a/2r_{\rm H}$ is the BH horizon angular frequency

Therefore, the output power is

$$P \sim \frac{1}{6\pi^2}\Phi^2\Omega_{\rm F}^2 \sim \frac{1}{24\pi^2}\Phi^2\Omega_{\rm H}^2$$

Credit: Sasha Tchekhovskoy

# Powering outflows: BZ mechanism




We are targeting jets, so $T^{\mu\nu} \approx T^{\mu\nu}_{\rm EM}$

In ideal MHD, the electric field vanishes in the fluid rest frame

i.e., $E \cdot B = 0$ or $^*F^{\mu\nu}F_{\mu\nu} = 0$

Therefore, in terms of the vector potential, we can write the ideal MHD condition as

$$A_{\phi,\theta}A_{t,r} - A_{t,\theta}A_{\phi,r} = 0, \quad \text{or,} \quad \frac{A_{t,\theta}}{A_{\phi,\theta}} = \frac{A_{t,r}}{A_{\phi,r}} \equiv -\omega(r,\ \theta)$$

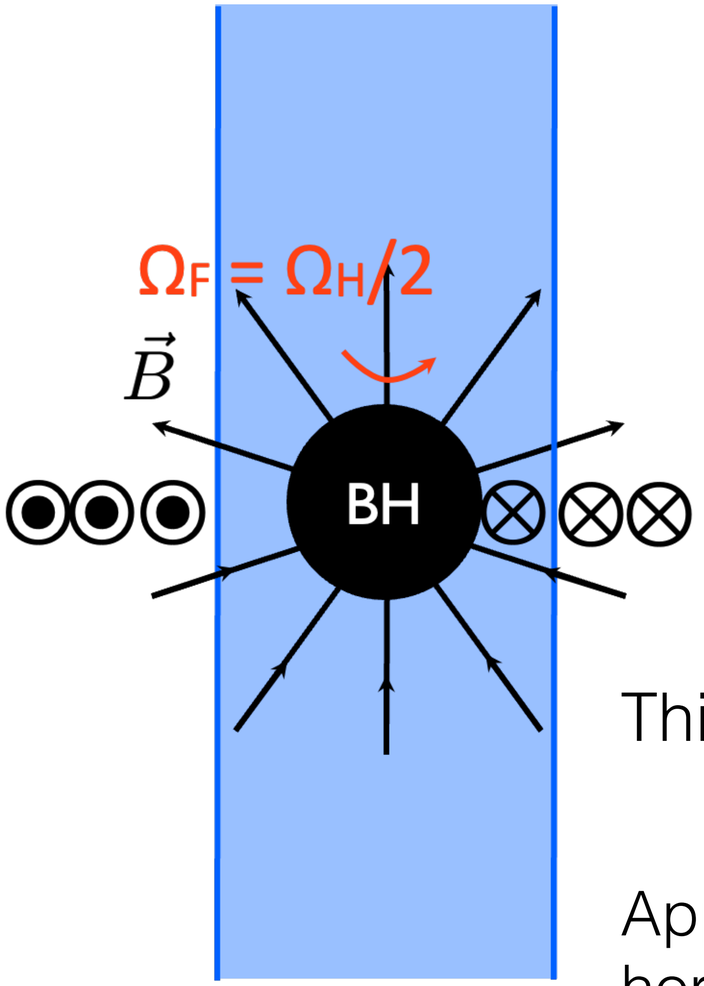
Fieldline rotation frequency

So, the radial energy flux is

$$F_E^{(\rm EM)} = -2(B^r)^2 \omega r \left(\omega - \frac{a}{2r}\right) \sin^2\theta - B^r B^\phi \omega \Delta \sin^2\theta.$$

McKinney & Gammie 04

# Powering outflows: BZ mechanism




$$F_E^{(\mathrm{EM})} = -2(B^r)^2 \omega r \left(\omega - \frac{a}{2r}\right) \sin^2\theta - B^r B^\phi \omega \Delta \sin^2\theta.$$

At the event horizon ($r = r_{\mathrm{H}}$), $\Delta = 0, \Omega_{\mathrm{H}} = a/2r_{\mathrm{H}}$

$$F_{\mathrm{E}}^{(\mathrm{EM})}(r_{\mathrm{H}}) \propto \omega(\Omega_{\mathrm{H}} - \omega)$$

Interestingly, the field line rotates at $\omega = \Omega_{\mathrm{F}} \to \Omega_{\mathrm{H}}/2$, which gives maximal power

This is thought to be a result of the Michel's monopoly solution matching the Znajek regularity condition

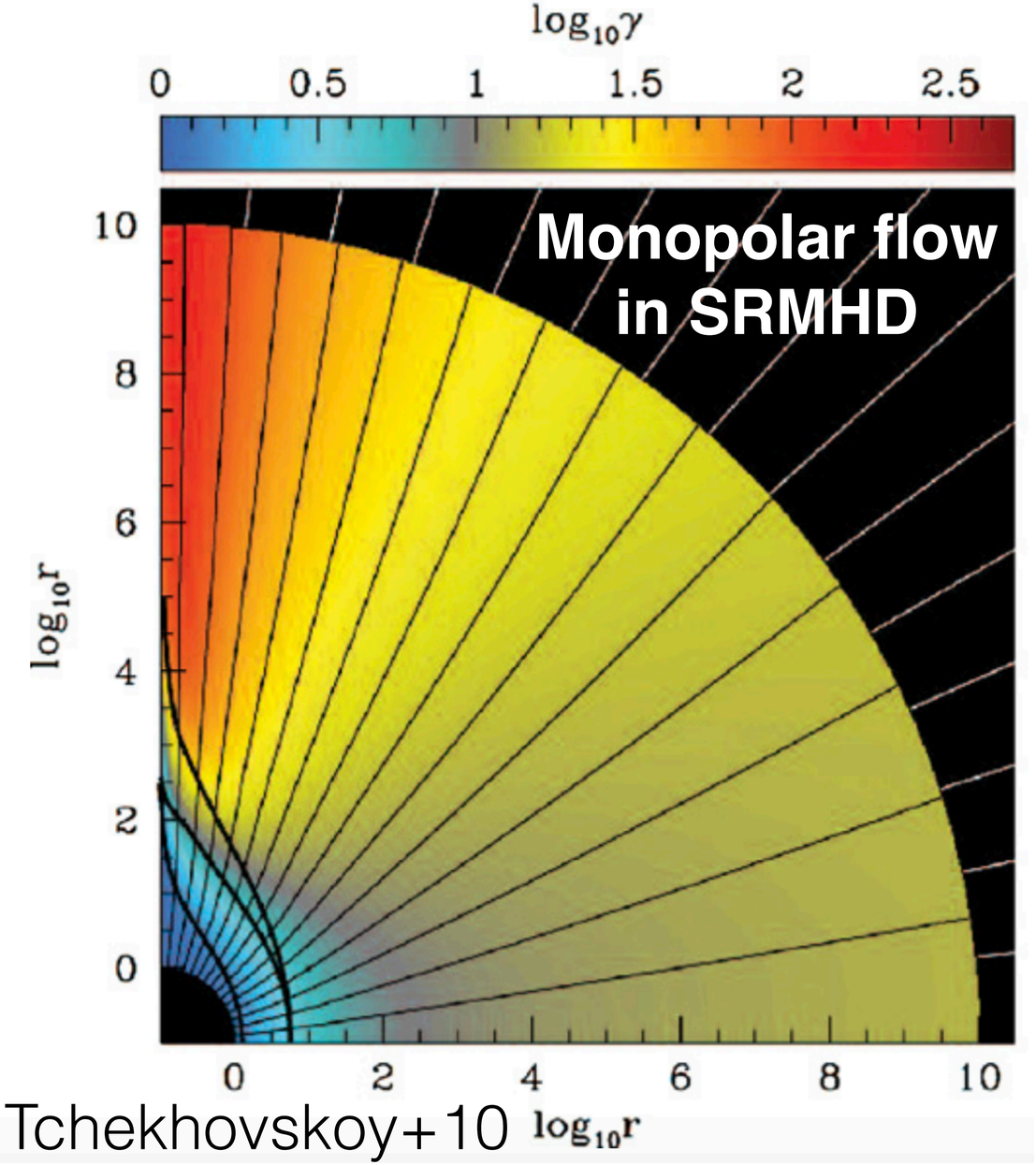
Applying Weber-Davis formalism to steady-state FF flow through the horizon, we get

$$B^\phi = \frac{\alpha H_\phi - B^r \sin^2\theta(2r\Omega - a)}{\Delta \sin^2\theta}$$

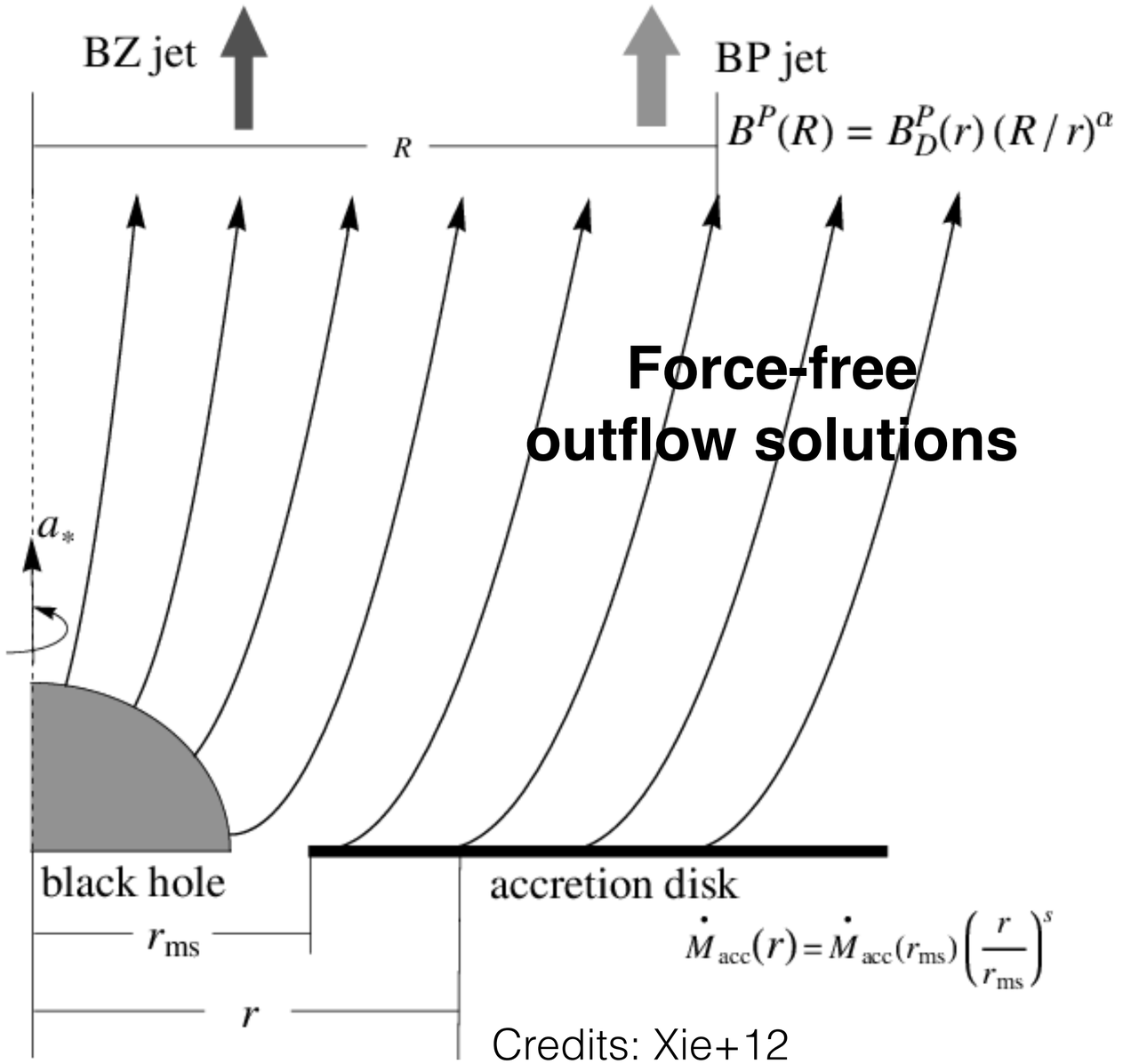
Critical point!

Komissarov 18,
McKinney & Gammie 04

# Jet Acceleration: launching paradigms

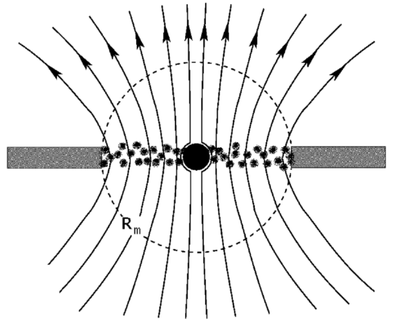



Tchekhovskoy+10

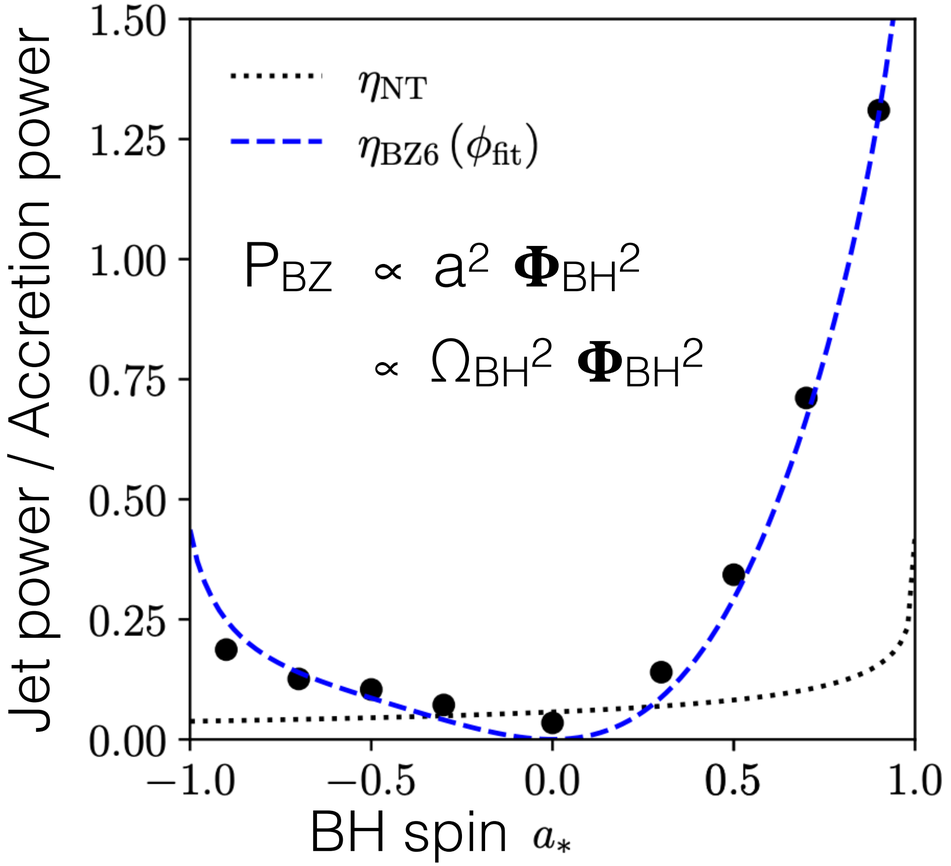



Credits: Xie+12

# Case Study: Jets from Magnetically arrested flows



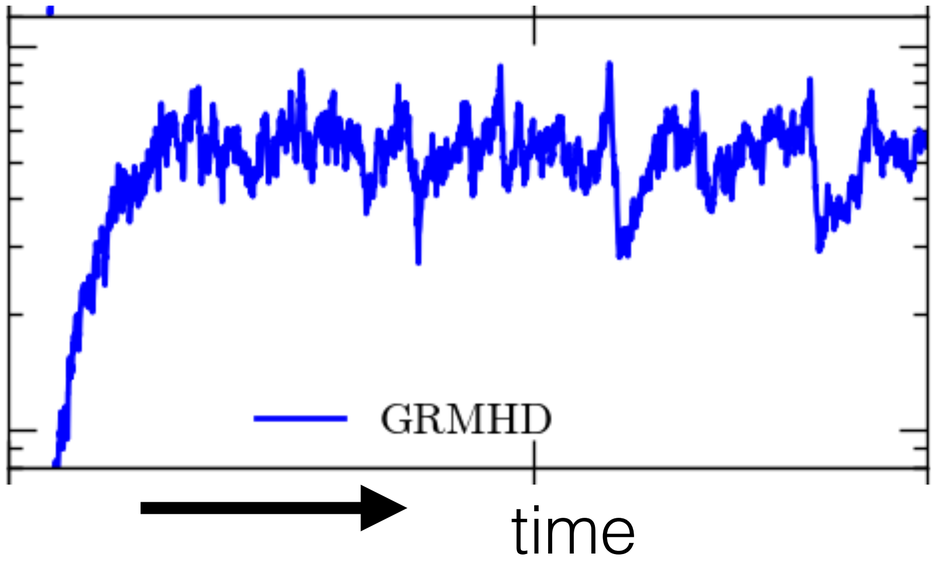

- MADs can have extremely efficient outflows: BH spin energy extraction (Tchekhovskoy+11) via Blandford-Znajek (BZ) mechanism
- Results in spindown: perhaps important for GRBs & ULXs (Narayan+17,+22, Ricarte+23, Lowell+23)
- Huge observational value: direct connection from accretion rates to jet power



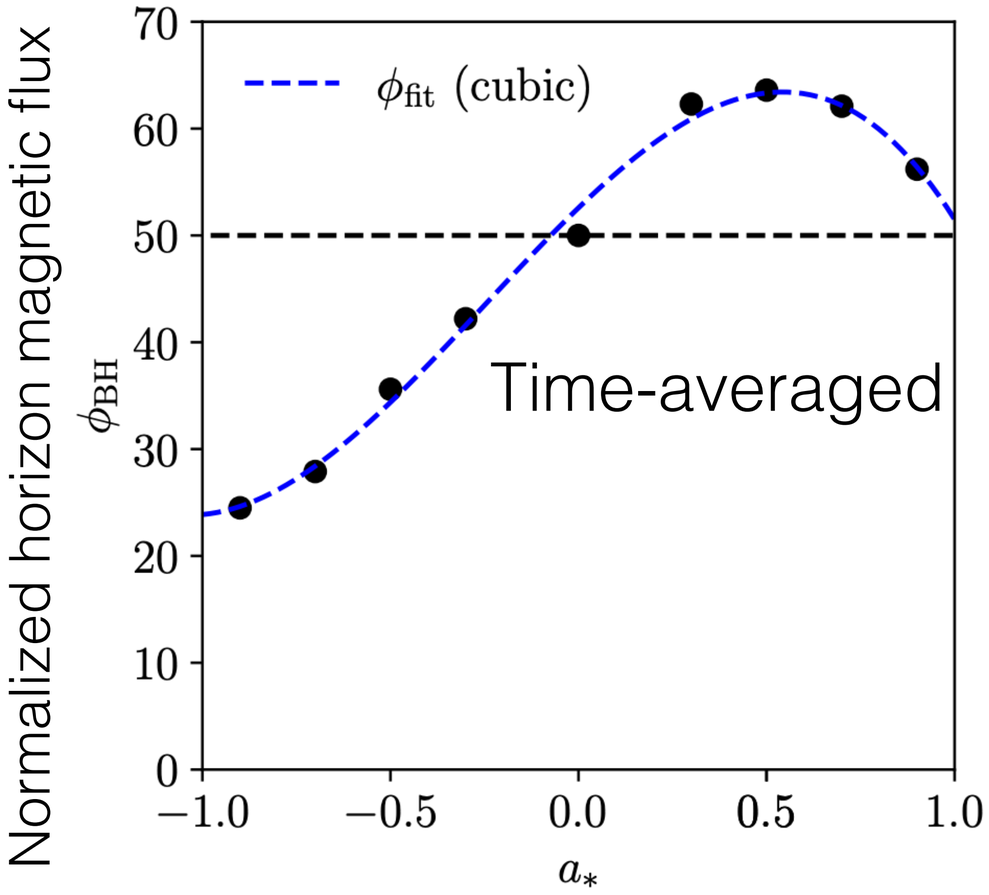

Tchekhovskoy+11,12
Narayan, Chael, **KC**, Ricarte & Curd 2022

# Magnetically arrested flows: magnetic flux

How does the magnetic flux at the event horizon vary over time? What is the saturation value?

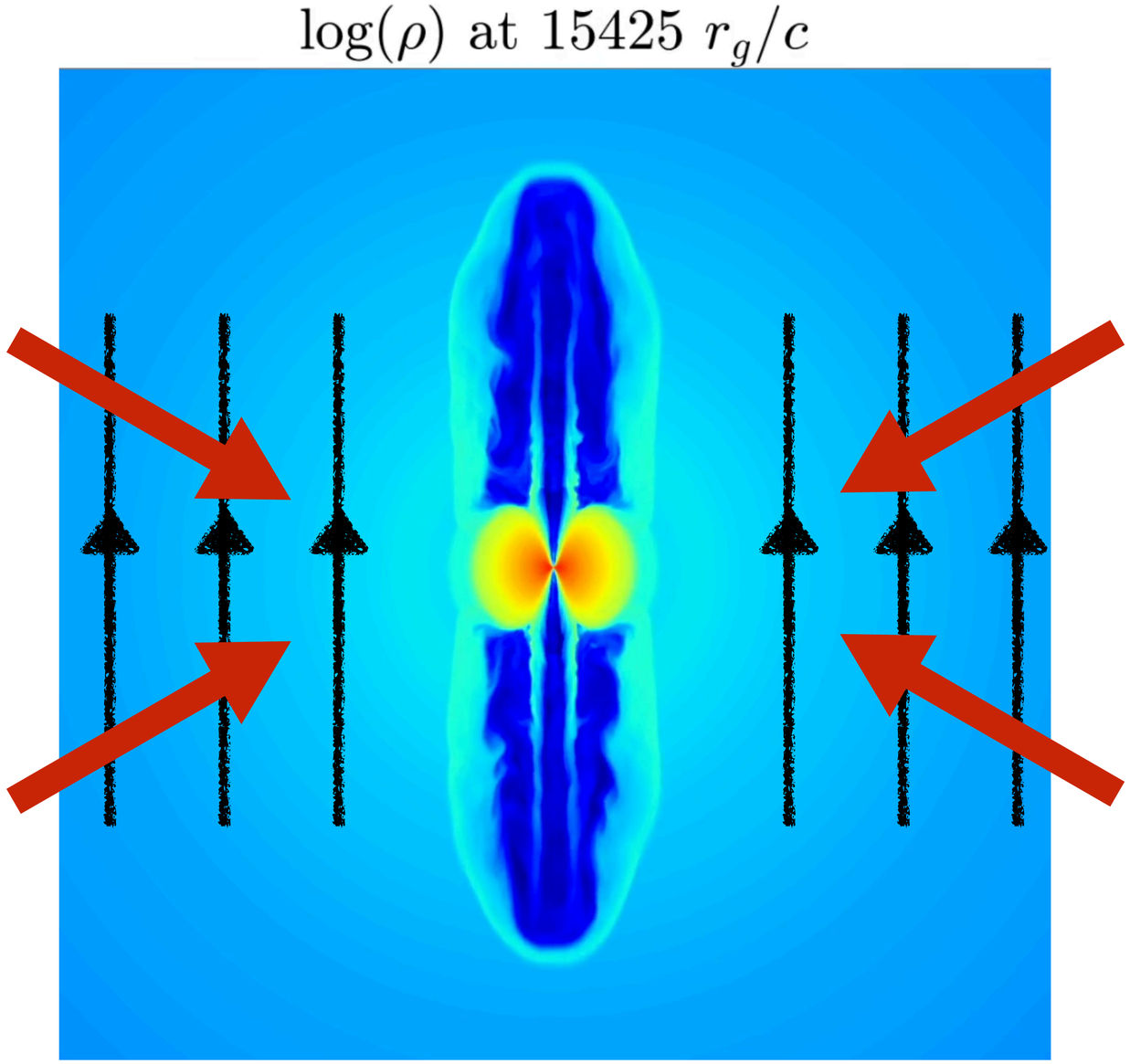



Time-variability of $\Phi_{BH}$




Tchekhovskoy+11,12
Narayan, Chael, **KC**, Ricarte & Curd 2022

# Magnetically arrested flows: magnetic flux




*Dynamo turbulence in the disk / interstellar medium? Larger scales?*

*Vertical Magnetic Field advection?*

*How is flux lost during state transitions?*

# Energetics along a flow

$\omega \equiv \nabla \times v$ Vorticity; related to fluid angular velocity

Using $v \times \omega = v \times (\nabla \times v) = \frac{1}{2}\nabla v^2 - (v\cdot\nabla)v$ and $\frac{dv}{dt} \equiv \frac{\partial v}{\partial t} + (v\cdot\nabla)v$

From Euler's equation, $\frac{dv}{dt} = -\frac{\nabla P}{\rho} - \nabla\Phi_g \rightarrow \frac{\partial v}{\partial t} + \nabla(\frac{1}{2}v^2 + \Phi_g) + \frac{\nabla P}{\rho} - v \times \omega = 0$

Using the thermodynamic relation $dU = T\,ds - P\,d(\frac{1}{\rho})$ & enthalpy, $h = U + P/\rho \rightarrow dh = T\,ds + \frac{dP}{\rho}$

Assume steady-state and constant entropy along a flow $(\partial/\partial t \rightarrow 0)$

$\frac{ds}{dt} = (v\cdot\nabla)s = 0 \rightarrow (v\cdot\nabla)P = \rho(v\cdot\nabla)h$

# Energetics along a flow

Dotting $\frac{\partial v}{\partial t} + \nabla(\frac{1}{2}v^2 + \Phi_g) + \frac{\nabla P}{\rho} - v \times \omega = 0$ with velocity $v$ & using, $\frac{\partial v}{\partial t} = 0$

We get Bernoulli's theorem: $\frac{dB}{dt} = (v \cdot \nabla)B = 0,$ where, $B \equiv \frac{v^2}{2} + h + \Phi_g$

Bernoulli's theorem states that in a steady flow of an ideal fluid, the Bernoulli function B is conserved along streamlines

Related to specific energy flux $\mu = \frac{F_E}{F_M} = -\frac{T^r{}_t}{\rho u^r} = -u_t(1 + h + \frac{b^2}{\rho}) + \frac{b^r b_t}{\rho u^r}$

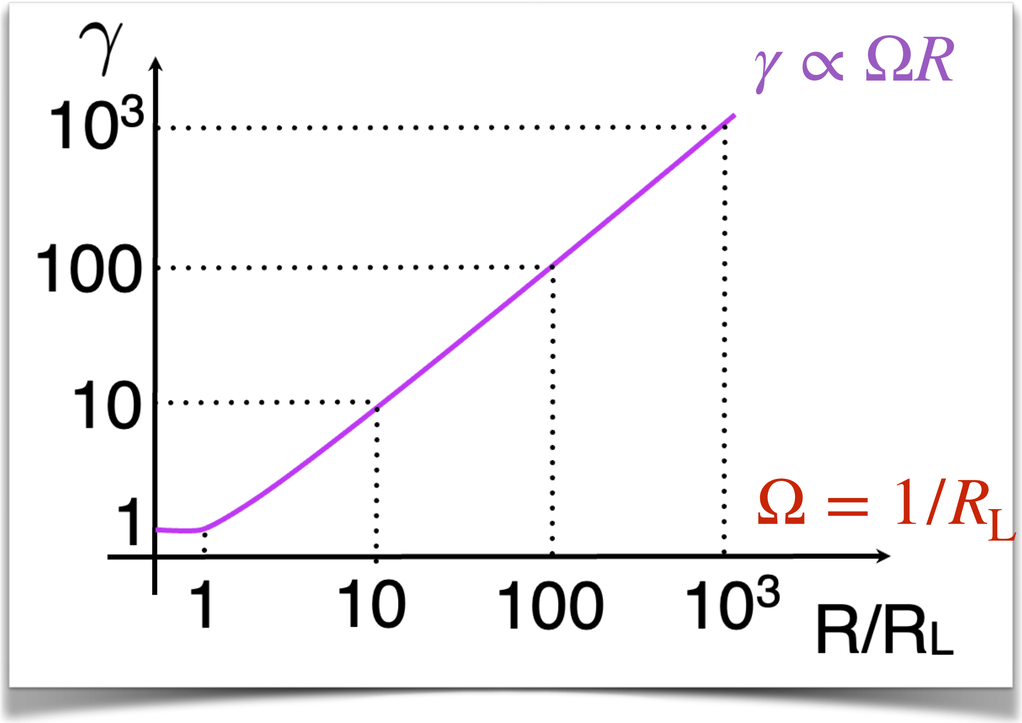
far from the black hole, $\mu \approx \gamma(1 + h + \sigma_{cold})$

# Jet Acceleration

Tchekhovskoy+09,+10

Assume force-free flow, drift velocity $v_{\rm d} = \dfrac{|E \times B|}{B^2} \sim \dfrac{E}{B}$

Lorentz factor (squared) $\gamma^2 = \dfrac{1}{1 - v_{\rm d}^2} = \dfrac{B^2}{B^2 - E^2}$ , where $|E| \approx |B_\varphi| \sim \Omega R B_r$




So, we have $\gamma^2 = 1 + \dfrac{B_\varphi^2}{B_r^2} = 1 + (\Omega R)^2$

Linear acceleration to high Lorentz factors

But assumes that $\sigma_{\rm cold}$ ( $= \dfrac{b^2}{\rho c^2}$ ) is very large

Credit: Sasha Tchekhovskoy

# Jet Acceleration

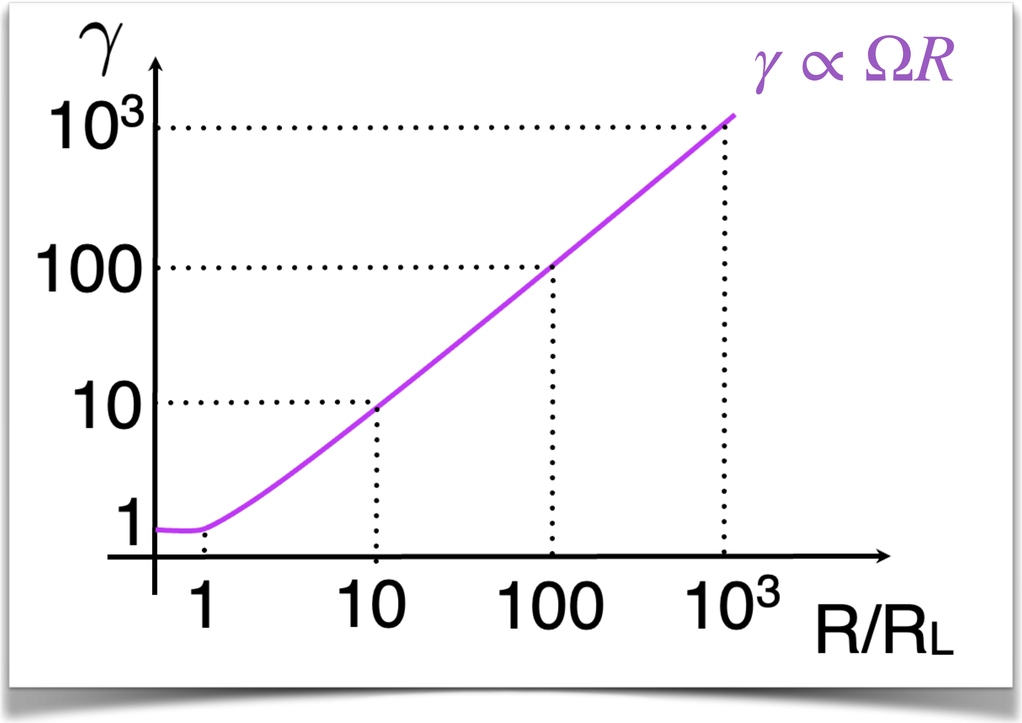
Force-free breaks down when the jet crosses the fast magnetosonic surface

We have $(\gamma v)_{\rm fast} = \sigma_{\rm cold}^{1/2}$ .

We can take $v_{\rm fast} \approx c = 1$ as we are in the relativistic regime

We know $\mu = \gamma(\sigma_{\rm cold} + 1)$, ignoring plasma enthalpy in the jet (cold flows)

$$\Rightarrow \mu \approx \gamma\sigma_{\rm cold} \Rightarrow \gamma_{\rm fast} \approx \mu^{1/3}$$




This is much lower than the asymptotic Lorentz factor $\gamma_{\rm max} \approx \mu$
(which occurs when all the available energies are transferred to kinetic)

Credit: Sasha Tchekhovskoy

# Jet Acceleration

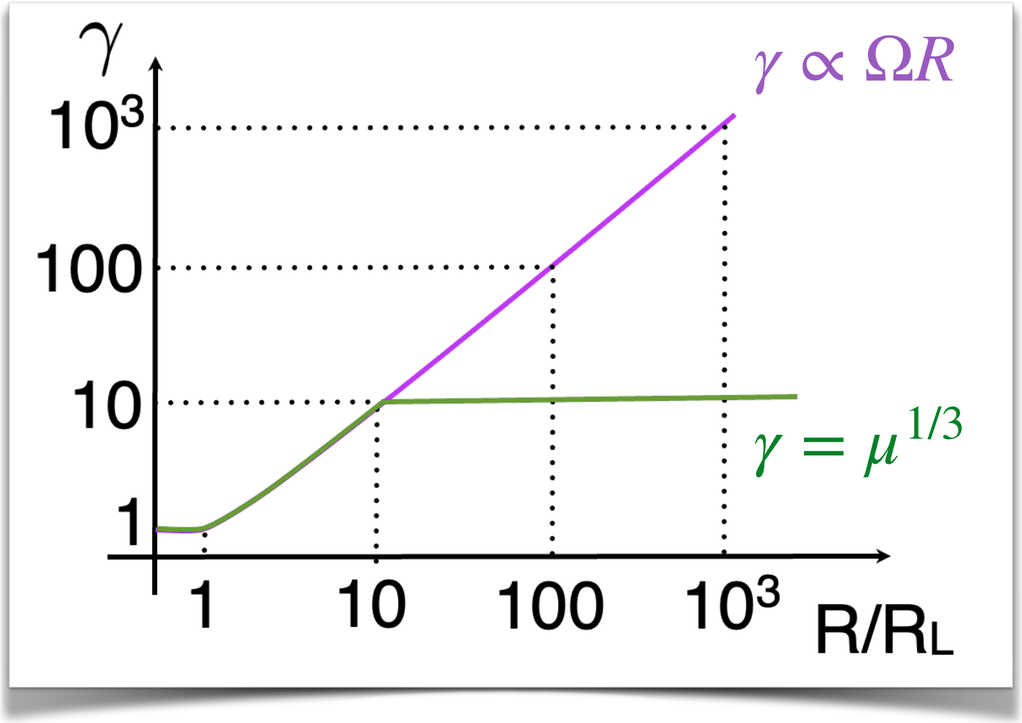
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# Jet Acceleration

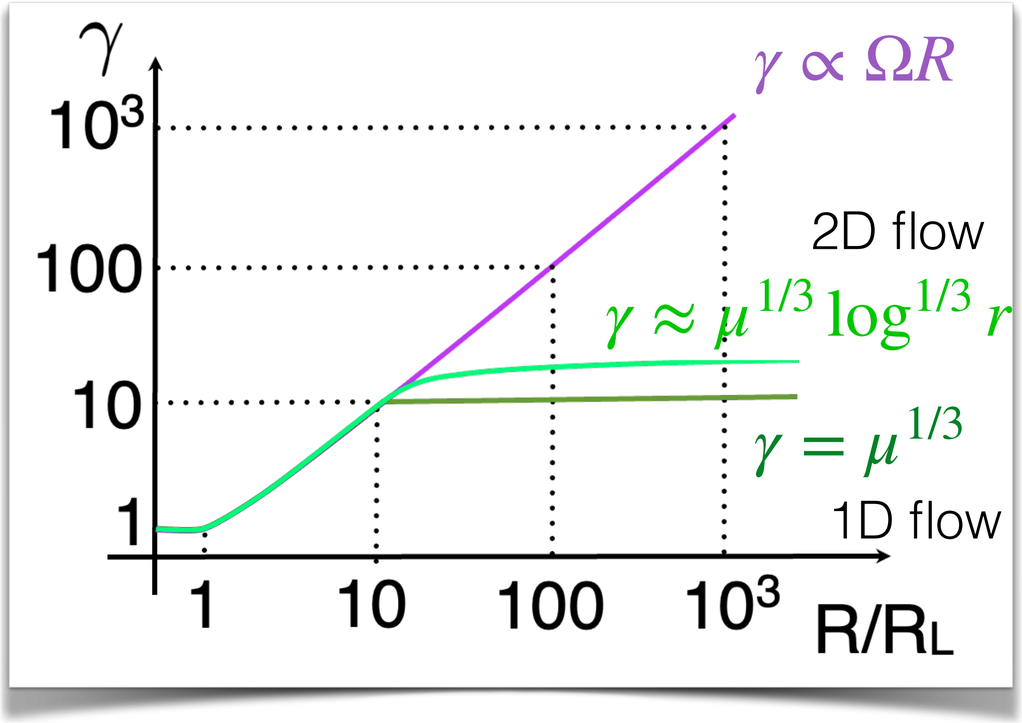
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This is much lower than the asymptotic Lorentz factor $\gamma_{\rm max} \approx \mu$ (which occurs when all the available energies are transferred to kinetic)

Logarithmic acceleration arises from curvature in magnetic field lines (Remove assumption that $B_\varphi^2 - E^2 \ll B_r^2$)

Credit: Sasha Tchekhovskoy

# Jet Acceleration: closely related to collimation

The specific energy flux can be rewritten using the enclosed magnetic flux

$$\Phi \equiv (\pi R_{\rm jet}^2 B_p)_{\rm fast}$$

We know EM energy flux

$$F_{\rm EM} = \frac{cEB_\varphi}{4\pi} = \frac{\Omega^2 R^2 B_p^2}{4\pi c}$$

The ratio of poloidal magnetic flux to rest mass flux is conserved

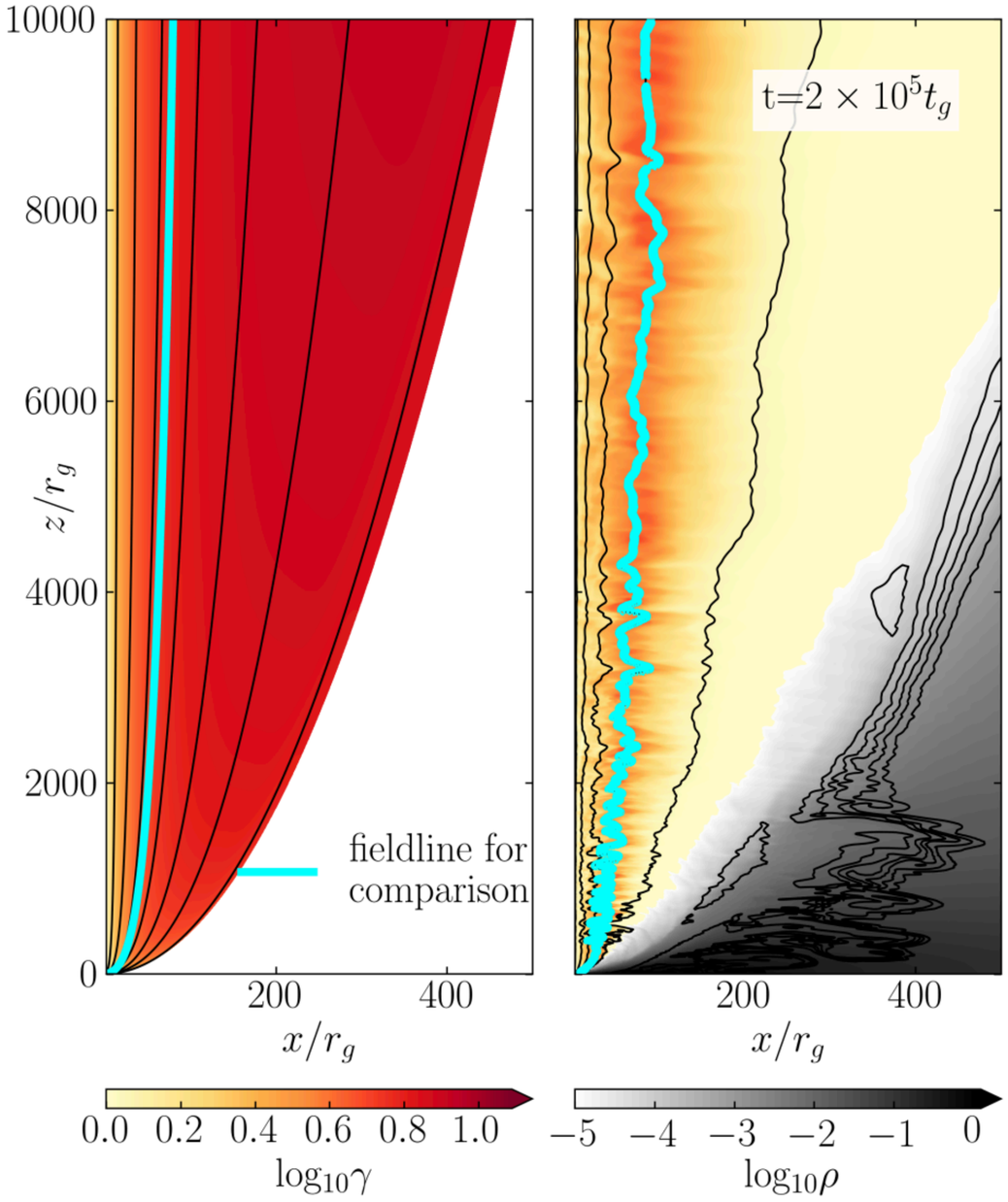
$$\gamma \rho v_p = \eta B_p$$

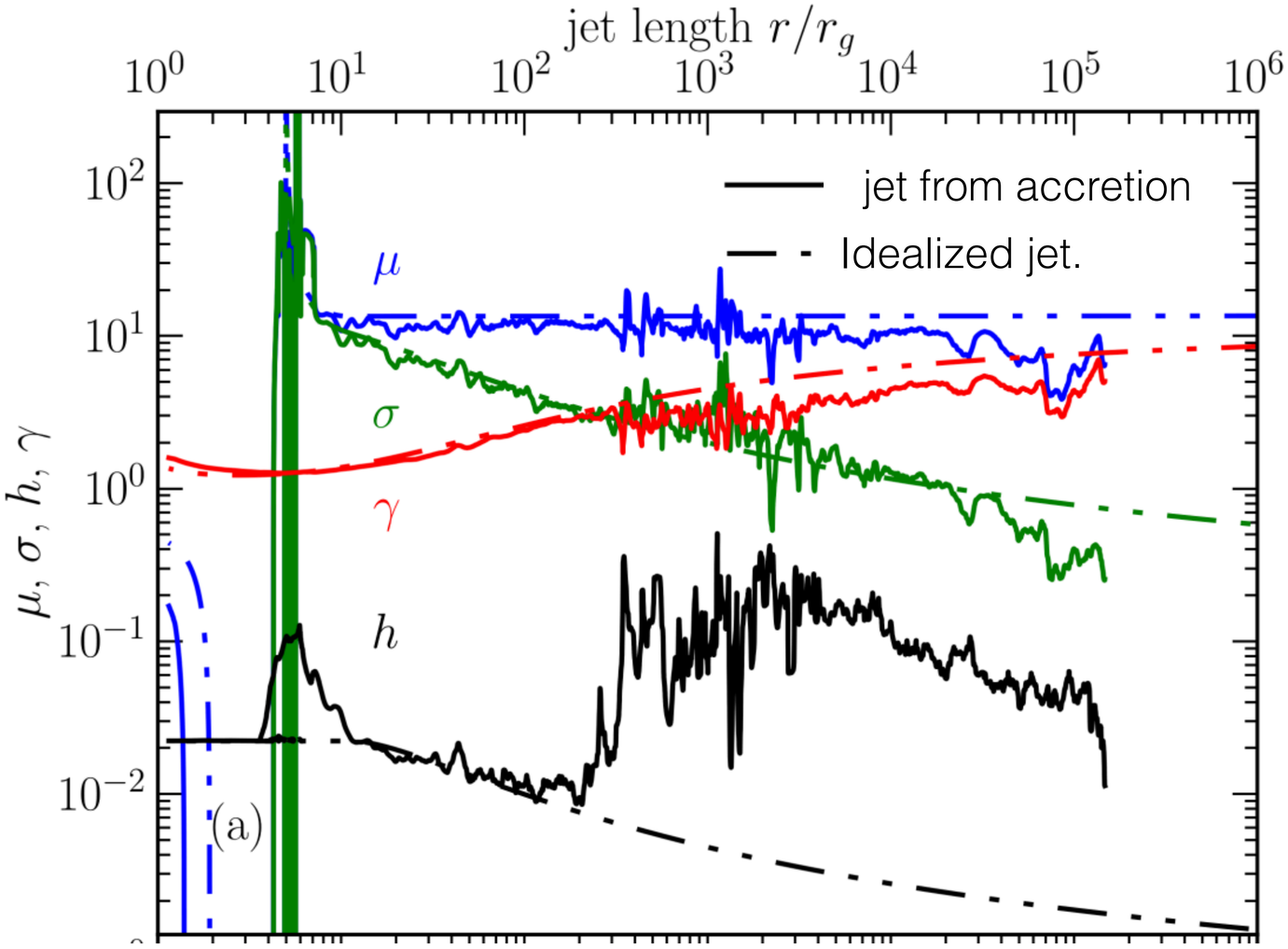
$$\text{So, } \mu = \frac{F_{\rm EM} + F_{\rm KE}}{F_{\rm M}} = \frac{F_{\rm EM}}{F_{\rm M}} + \gamma \Rightarrow \frac{\gamma}{\mu} = 1 - \frac{\pi R^2 B_p}{\Phi}$$

We see differential bunching of field lines in accelerating jets

Conical jets accelerate quickly but relatively inefficiently w.r.t. parabolic core

# Jet Acceleration (simulation)




$$\mu \approx \gamma(1 + h + \sigma_{\rm cold})$$

KC+19

# Jet Acceleration in different systems

Simulations of magnetized confined jets:

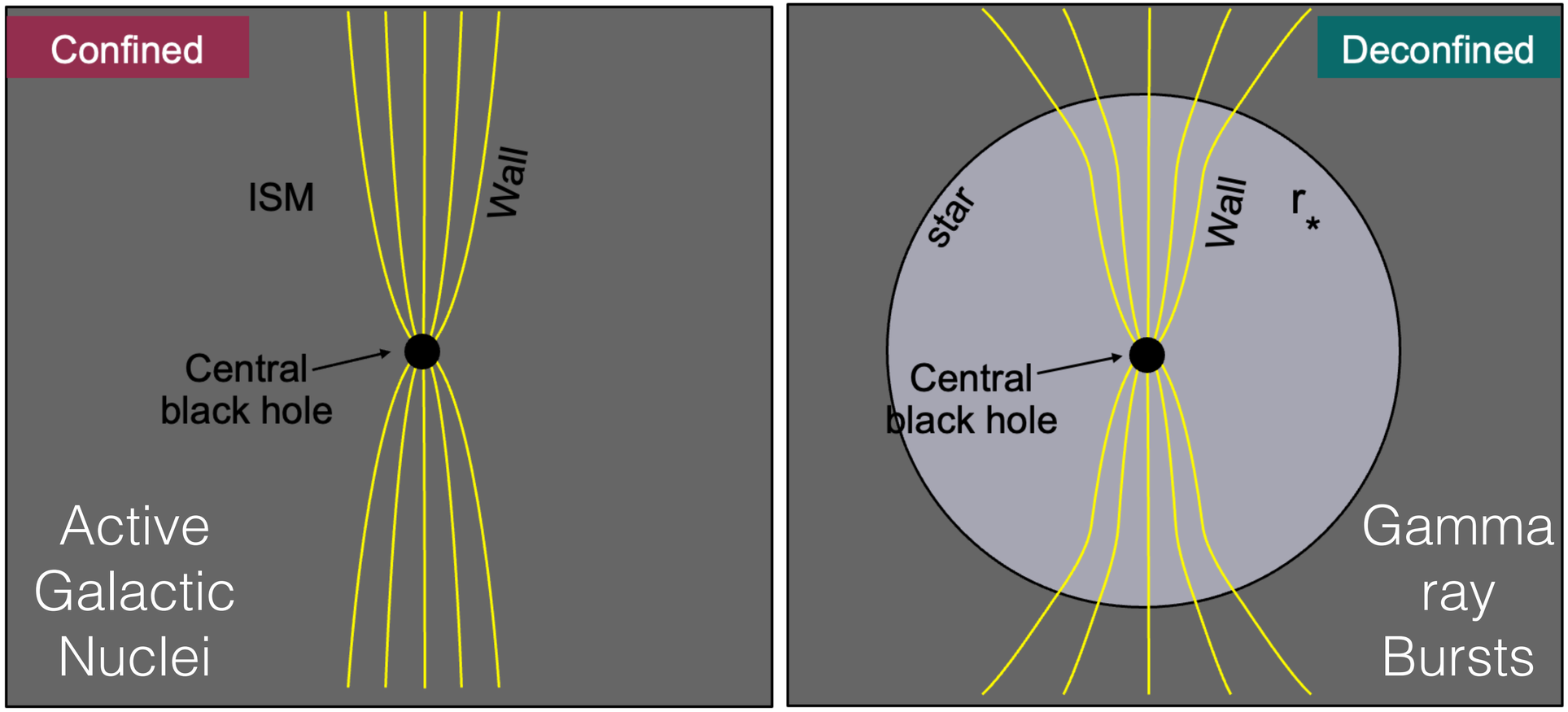
Parabolic acceleration

(Komissarov et al., MNRAS, 2009)

GRB jets are DEconfined:

Rarified acceleration

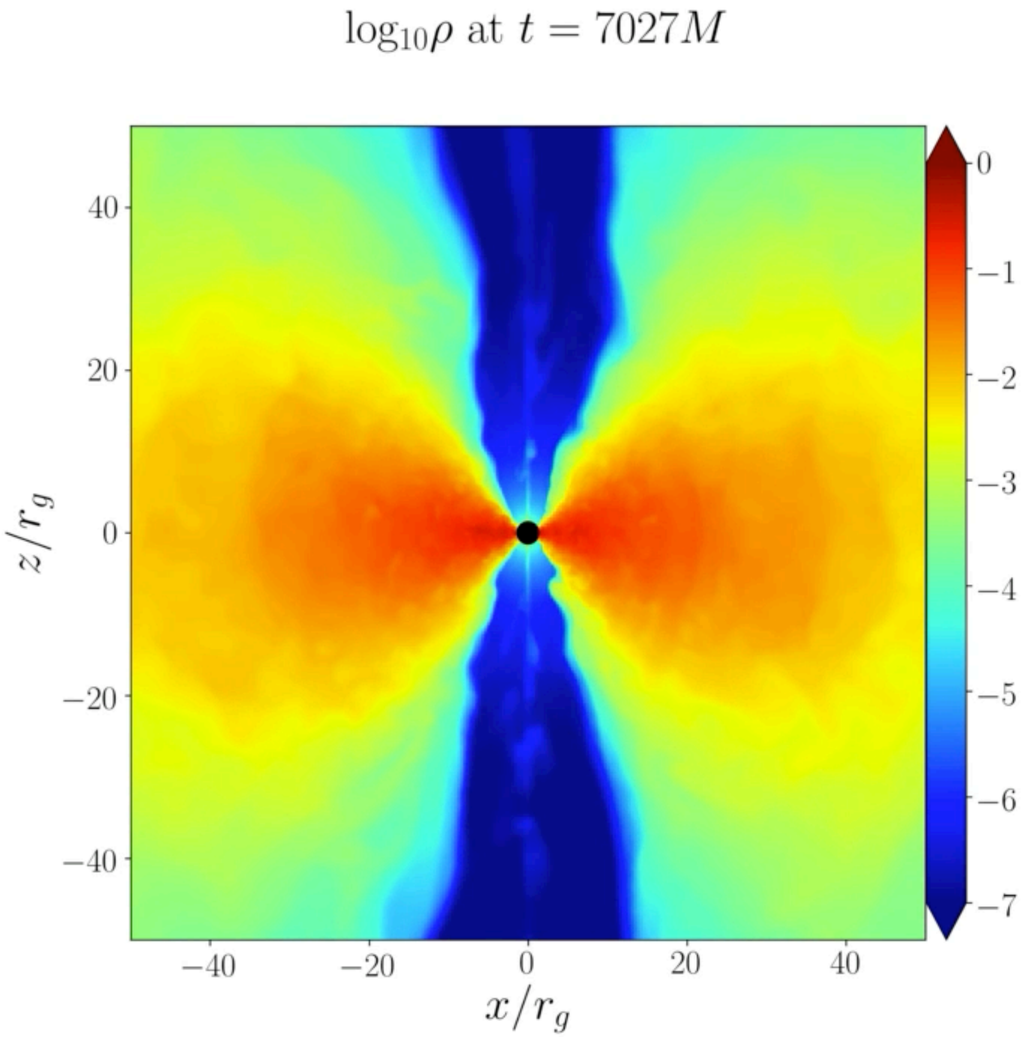
(Tchekhovskoy, Narayan, McKinney, New Astronomy, 2010)




Credit: Sasha Tchekhovskoy

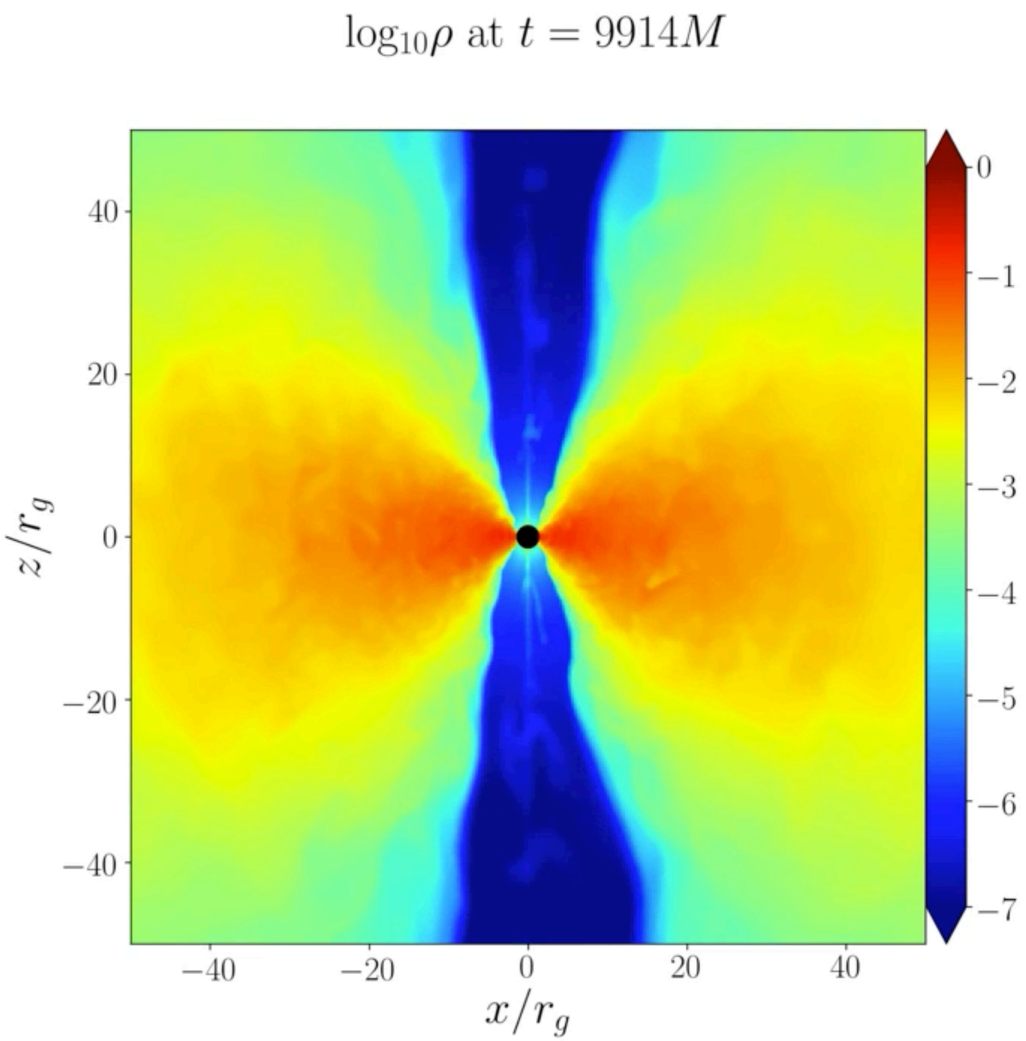
# Relativistic jets in GRMHD

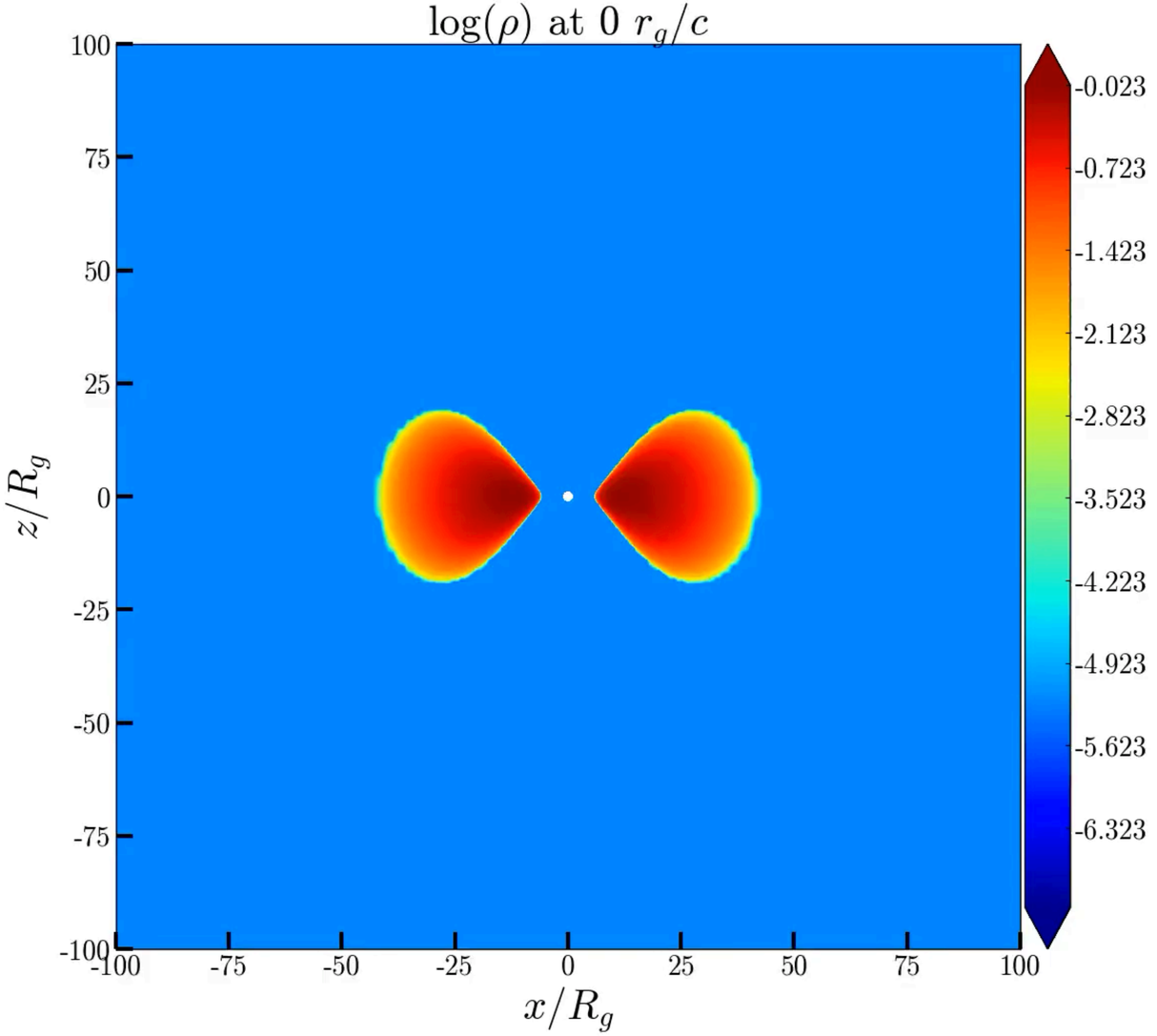
*Launching, Morphology and Acceleration*



# Relativistic jets in GRMHD

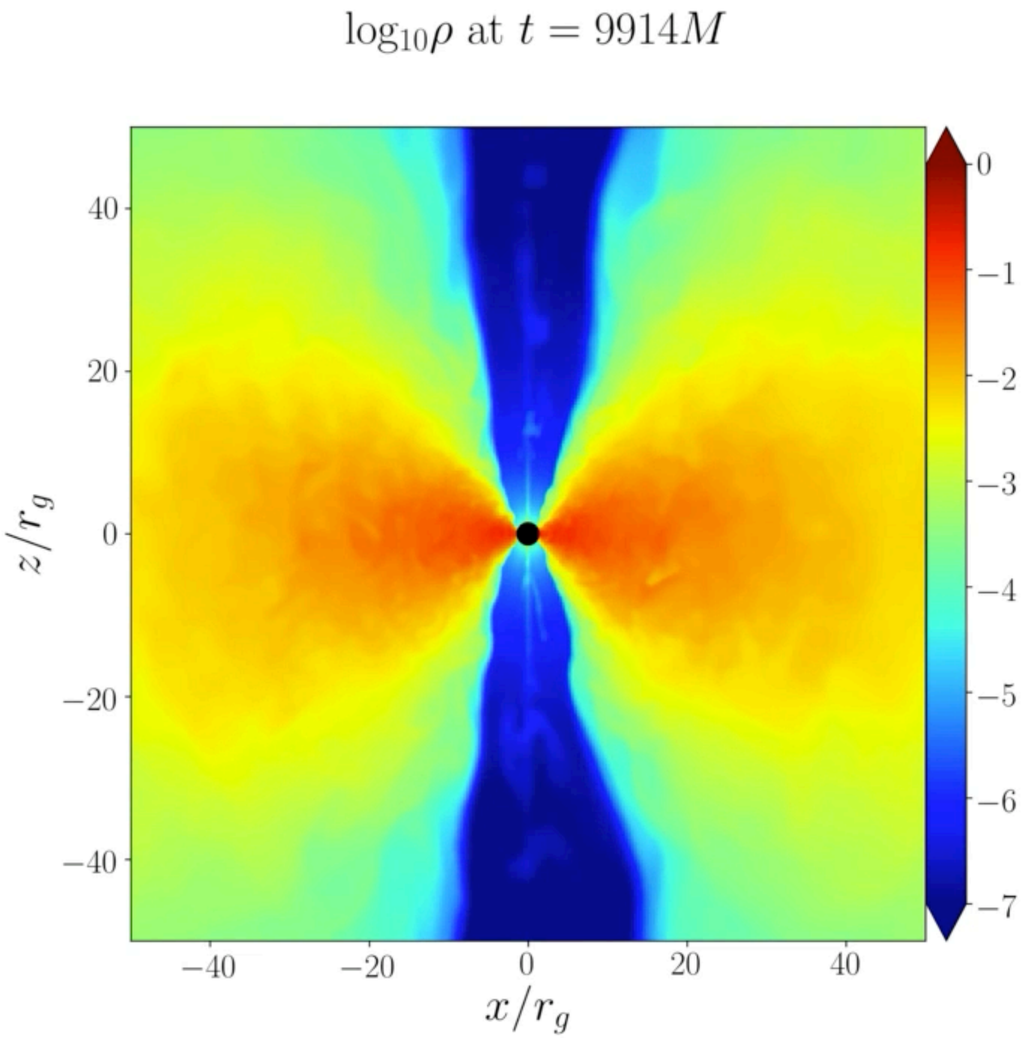
*Launching, Morphology and Acceleration*

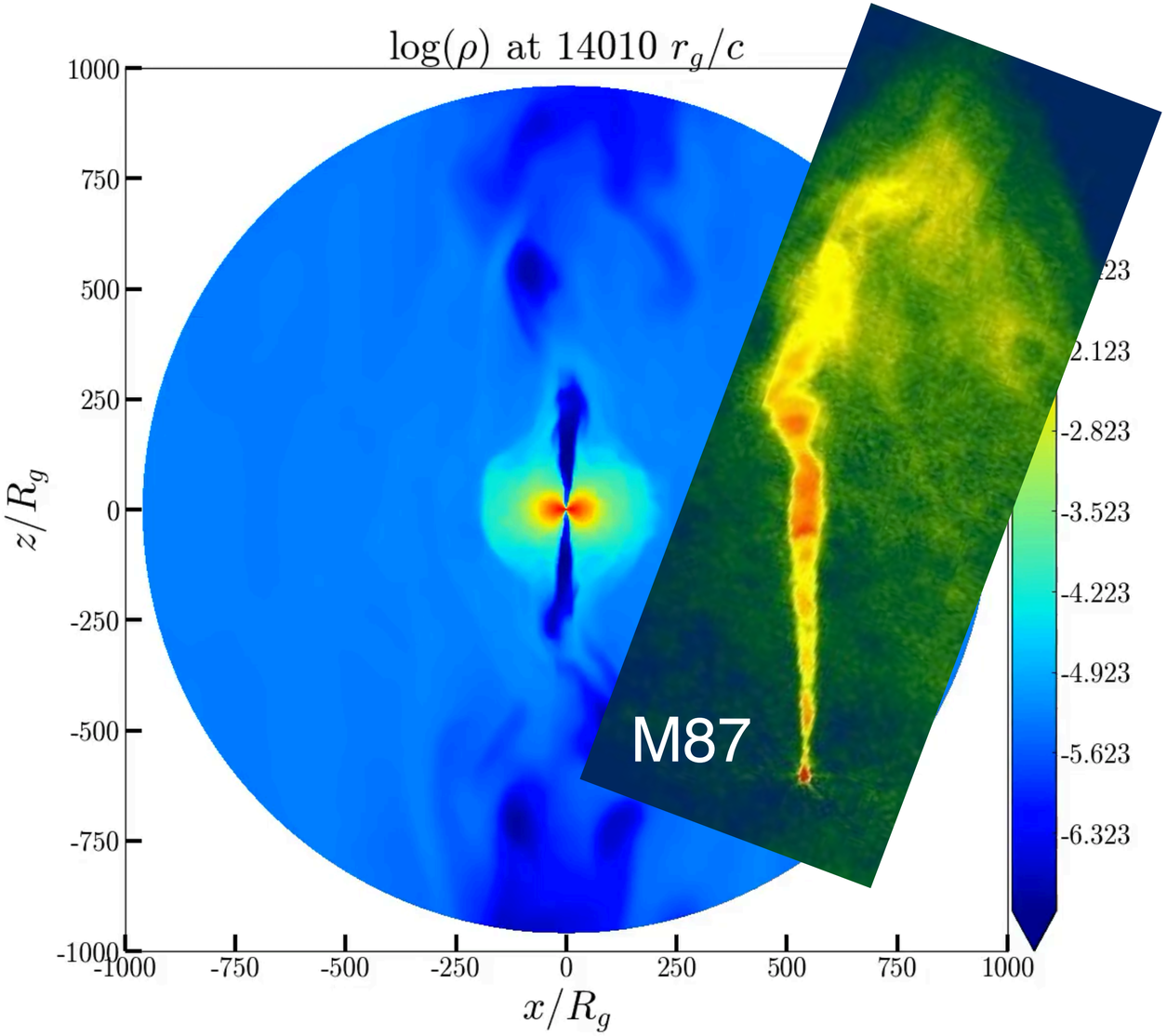


# Relativistic jets in GRMHD

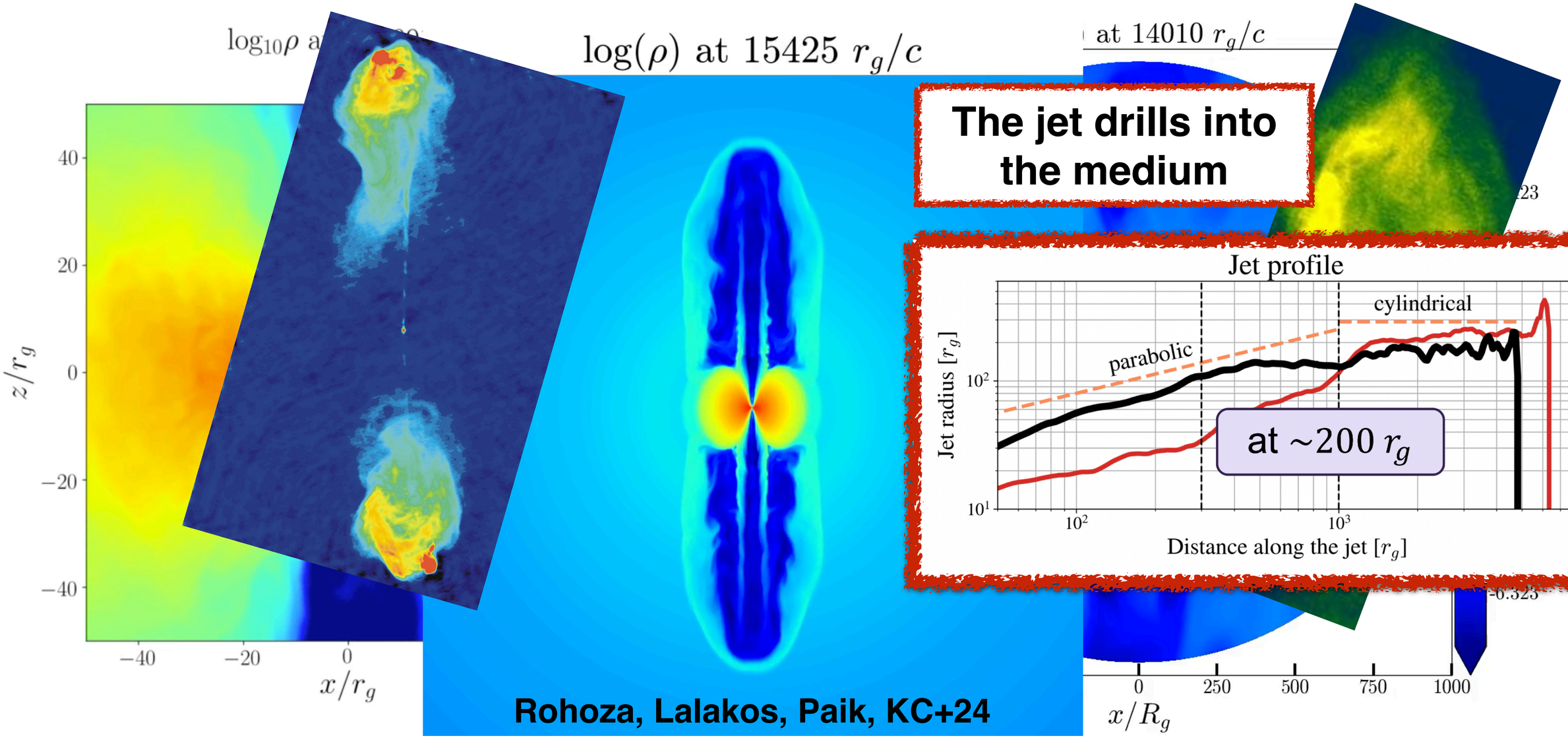
*Launching, Morphology and Acceleration*



# Relativistic jets in GRMHD

*Strong magnetic fields stabilize jets*




**The jet drills into the medium**



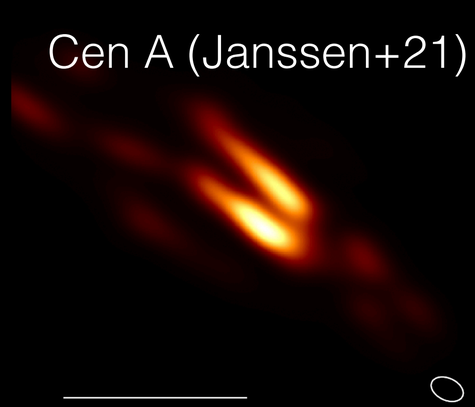

**Rohoza, Lalakos, Paik, KC+24**

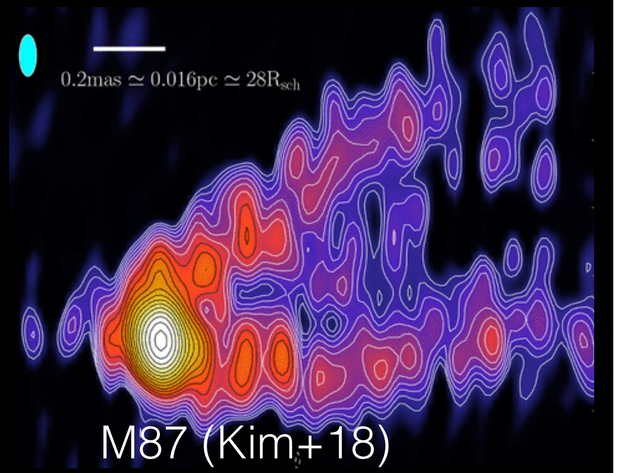
**Lalakos,..,KC+23**

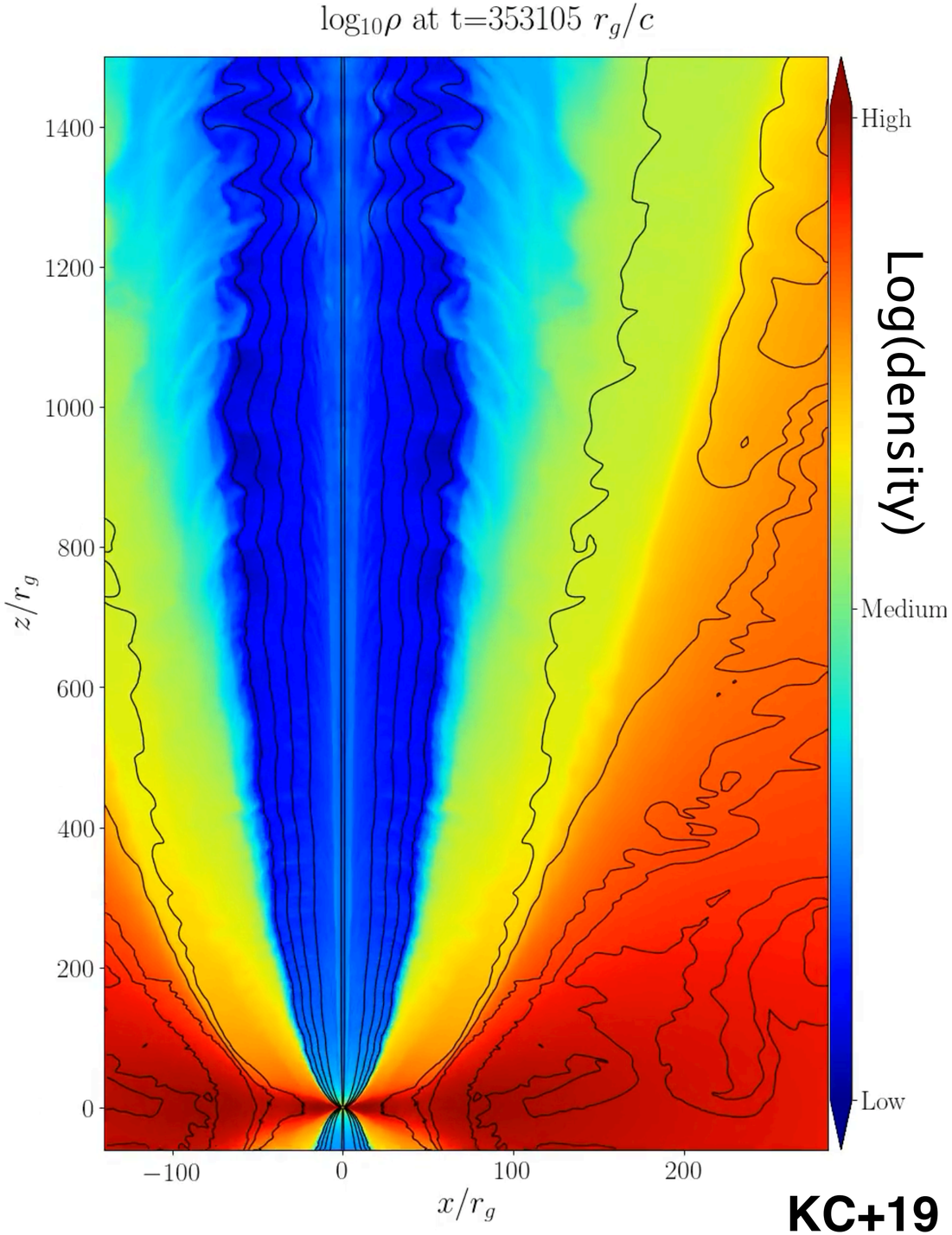
# Jet boundary instabilities

The 2D jet accelerates
- pinches, KH, shearing
- mass-loads
- decelerates!

**Limb-brightening in AGN jets**

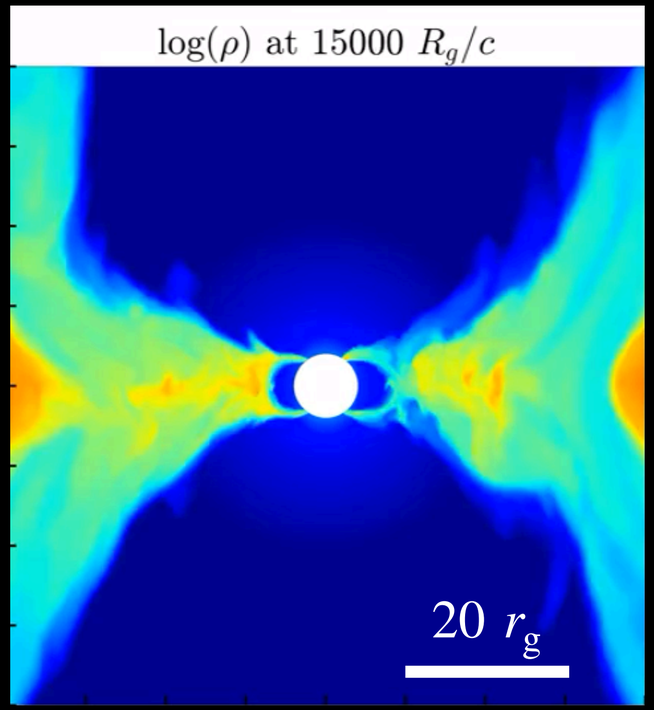


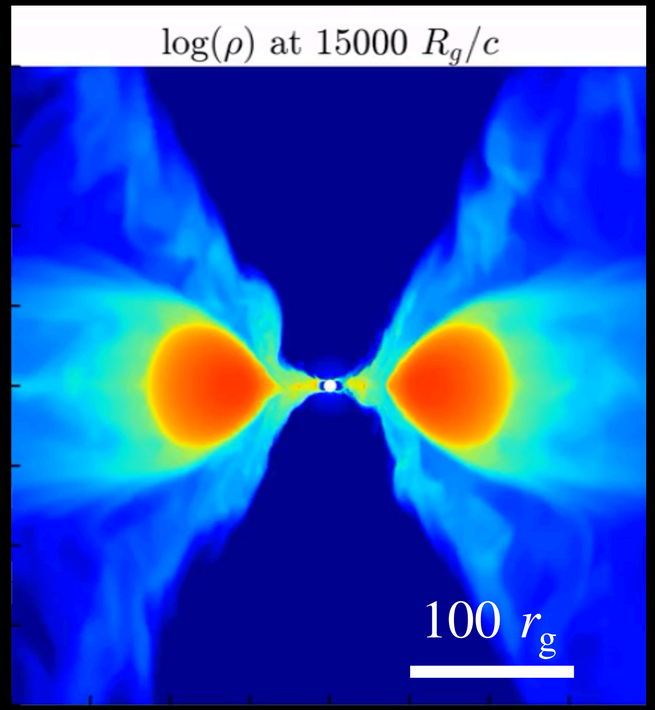




KC+19

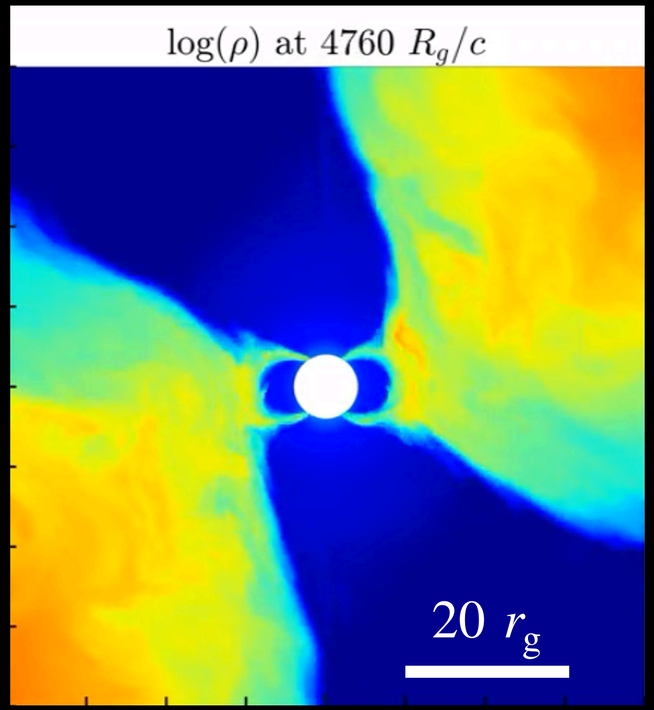
# Fun things to do…..accreting pulsars and jets

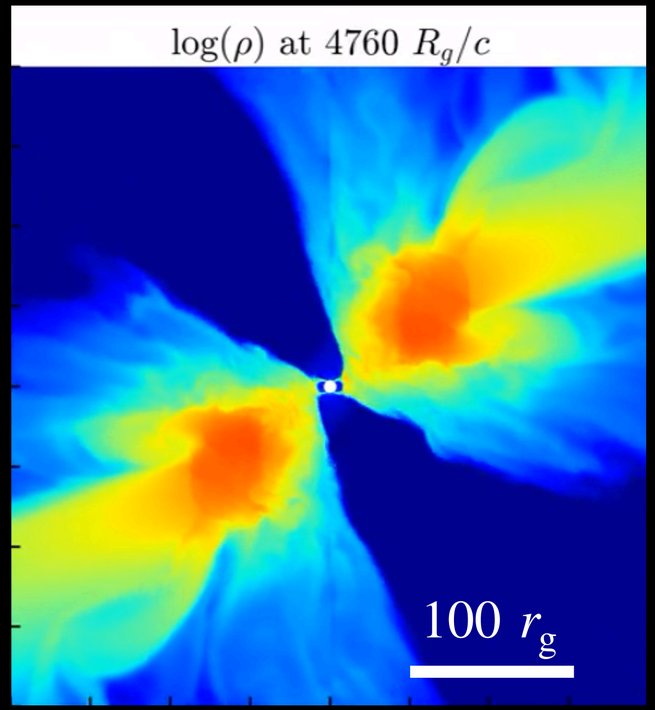



- Tracking accretion through columns
- Feedback on accretion from NS wind and pulsar current sheet

# Misaligned Accretion onto Pulsars

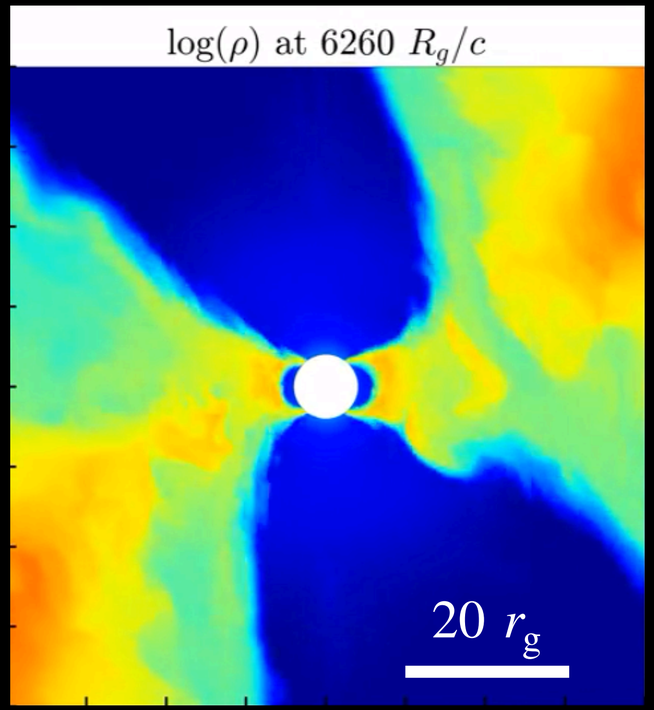


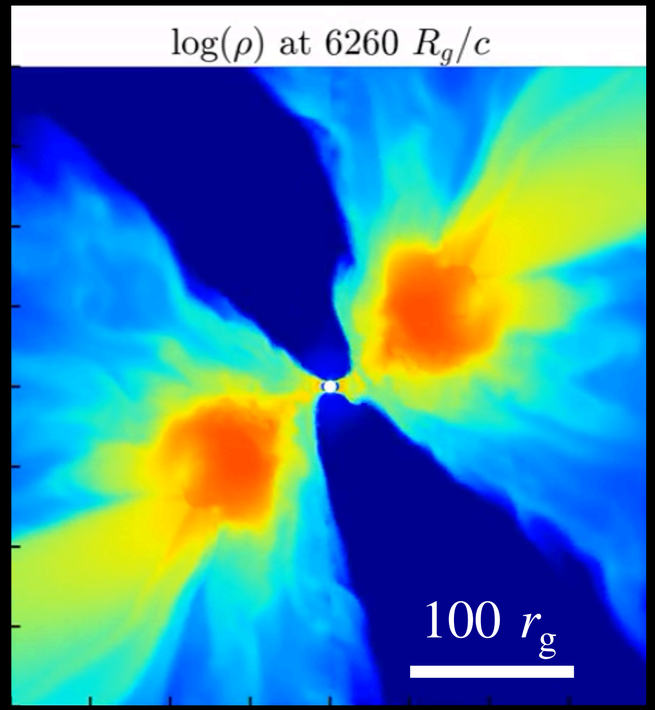




- Preliminary simulations indicate
  - Disk alignment due to strong NS field
  - Disk breaking can occur

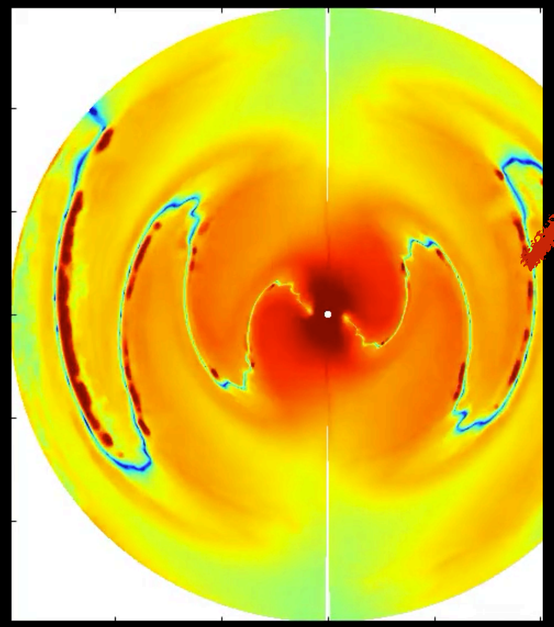
*KC+ (in prep)*

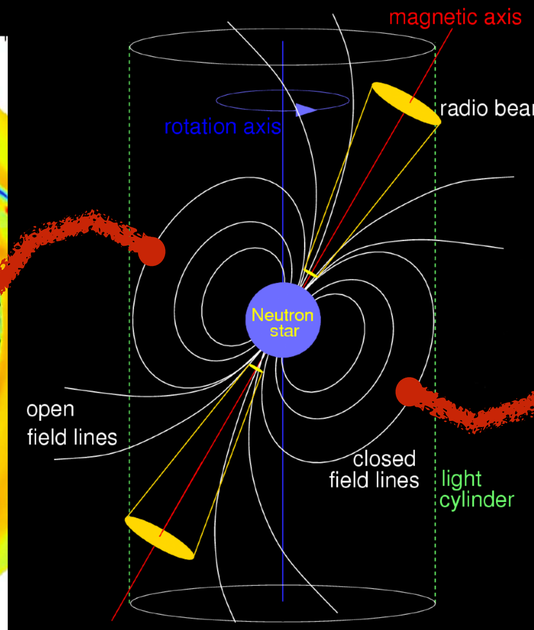
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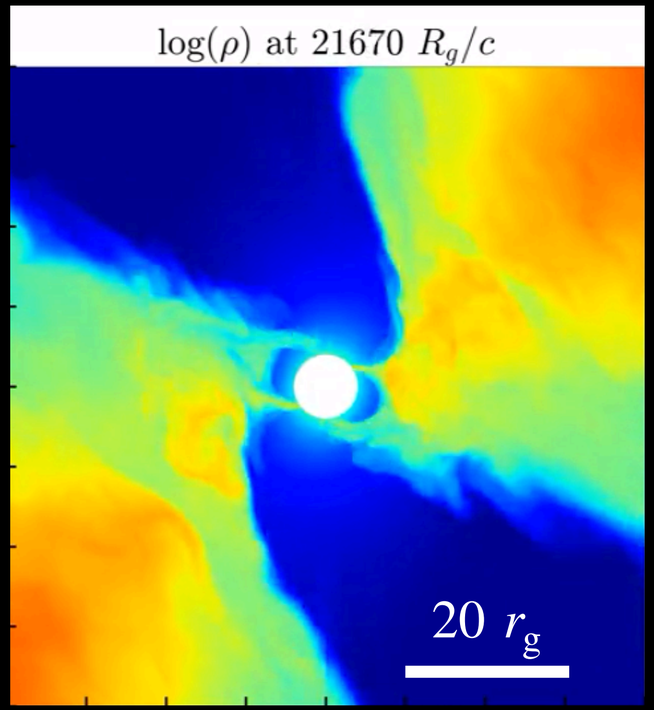


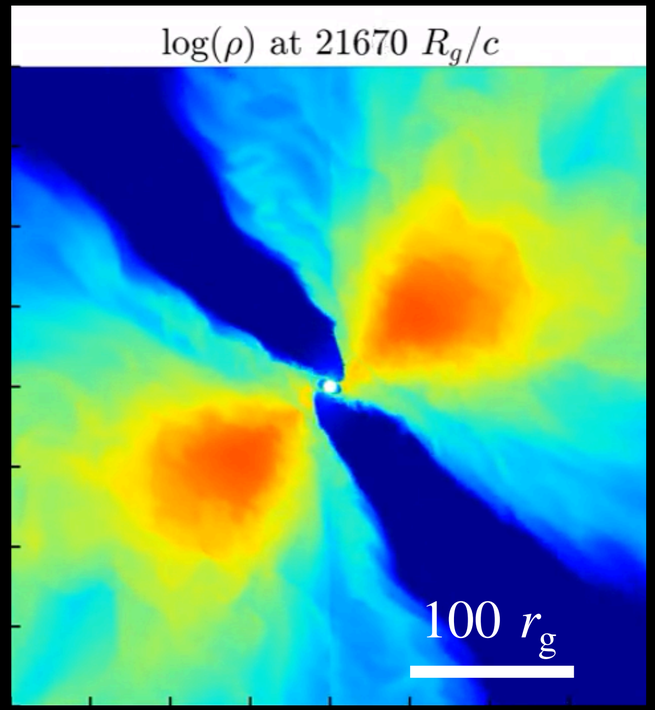



*Obliquity in pulsars increases outward Poynting flux*

*KC+ (in prep)*

# Misaligned Accretion onto Pulsars: Obliquity

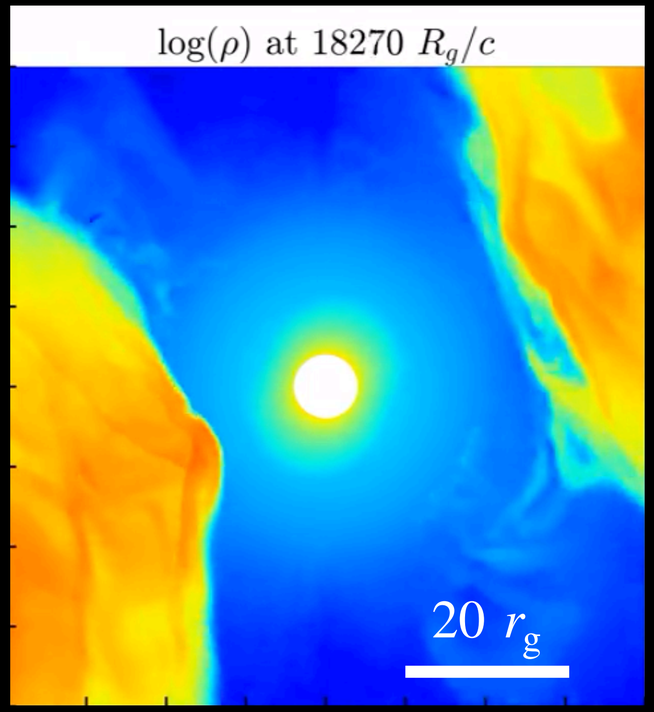


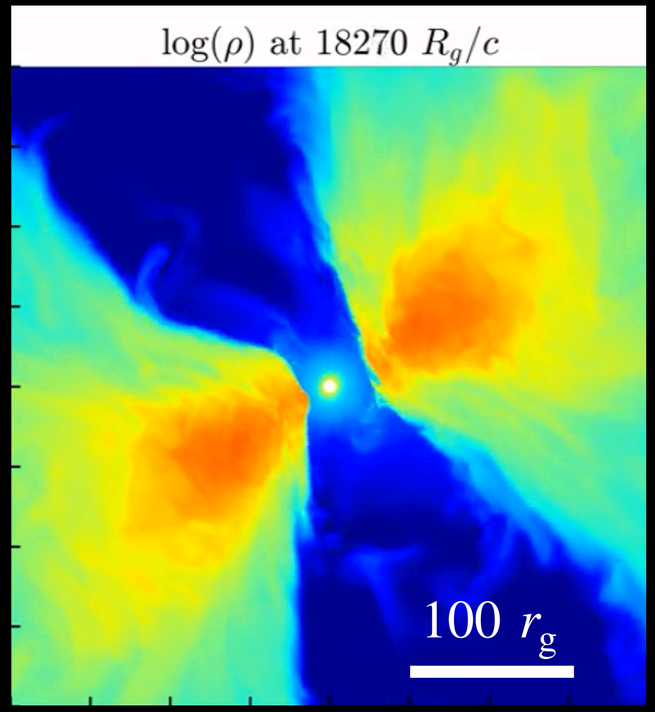




- Preliminary simulations indicate
  - Disk alignment due to strong NS field
  - Disk breaking can occur
  - Striped winds
  - Potential as a strong X-ray source

*KC+ (in prep)*

# Misaligned Accretion onto Pulsars: Current sheets




- Preliminary simulations indicate
  - Disk alignment due to strong NS field
  - Disk breaking can occur
  - Striped winds
  - Potential as a strong X-ray source
  - Current sheet whips the disk, stripping off gas

*KC+ (in prep)*